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Flows & Liquidity

The need for deleveraging

- The investor deleveraging phase that started in June appears to be still ongoing and we see more room for deleveraging in leveraged equity ETFs, options and margin accounts, thus acting as a headwind for equities going forward.
- The excessive growth of leveraged equity ETFs can be self-correcting: if excess leverage creates higher frequency of VaR shocks, a volatile range-trading in equities would over time shrink the AUM of leveraged equity ETFs.
- That said, it will perhaps take another three months or so of volatile range-trading in equities until the leveraged equity ETF AUM ratios to the market cap of underlying stocks return to pre-April levels for either memory stocks or all equity ETFs.
- Once this deleveraging abates, longer-term equity demand/supply dynamics should provide a support.
- Some encouraging signs from bitcoin futures.

Cross Asset Fund Flow Monitor

Current level shows the latest percentile of weekly flows; Min is denoted by 0 and Max by 1. As of 8th July 26.

MF & ETF Flows	Min	Max	4 wk avg (\$bn)	2025 avg (\$bn)
All Equities			29.6	8.1
All Bonds			16.8	11.3
US Equities			20.3	3.5
US Bonds			10.1	4.3
Non-US Equities			9.3	4.6
Non-US Bonds			6.7	7.0
US HG Bonds			10.1	3.4
US HY Bonds			0.7	0.4
US Lev. Loans *			0.3	0.0
US MMFs			-0.3	4.0
EM Equities			-1.3	0.9
EM Bonds			0.74	0.50
Japan Equities			0.8	-0.1
China Equities			-0.91	-0.15
Europe				
Europe Equities			-0.6	0.9
Europe Bonds			4.2	4.3
Europe HG Bonds			0.7	0.9
Europe HY Bonds			0.59	0.17
Europe MMFs			6.9	3.8
Other Equities			10.72	3.00

Source: Lipper, ICI, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

* US LL historical flows are monthly averages converted to weekly for comparison. China onshore A-share ETFs have been excluded.

Global Markets Strategy

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Cross Asset Positioning Monitor

Current level shows the latest percentile, Min is denoted by 0 and Max by 1.

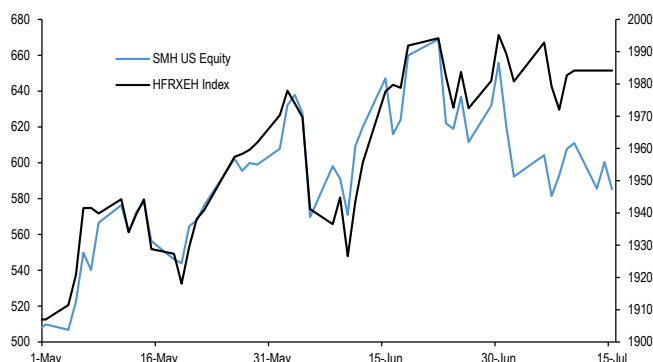
As of 14-Jul-26	MIN	MAX	Current percentile
Equities			0.64
Govt Bonds			0.64
Credit			0.29
Dollar			0.79
Commodities ex Gold			0.51
Gold			0.49
Bitcoin			0.56
EM Equities			0.81
EM Bonds/FX			0.22
Japan Equities			0.72
Europe Equities			0.87
US Equity Sectors:			
Energy			0.74
Materials			0.51
Industrials			0.42
Discretionary			0.54
Staples			0.14
Health Care			0.43
Financials			0.22
Technology			0.65
Communication Services			0.35
Utilities			0.47

Source: J.P. Morgan Flows & Liquidity.

Cross Asset Positioning Monitor aggregates across the various position indicators of Appendix ranging from positioning proxies across various futures contracts, momentum signals as proxies of how trend-following funds/CTAs are positioned, mutual fund betas as proxies of how mutual fund managers are positioned, risk parity fund positioning and leverage proxies, hedge fund betas as proxies of how hedge fund managers are positioned, client surveys, asset allocation estimates of private non-bank investors at global level, short interest indicators, etc.

- Despite a decline in S&P500 and Nasdaq indices during June, the performance of equity-focused hedge funds such as Equity Long/Short and Equity Sector TMT was positive at 1.2% and 3.7%, respectively. Their performance was likely helped by semiconductors, with the SMH Semiconductor ETF up 9.5% in June vs. -14.5% loss for US hyperscalers on the month. In other words, last month's hedge fund performance is consistent with equity-focused hedge funds maintaining an overweight exposure to semiconductors through June.
- We do not have comprehensive data for July yet from monthly reporting hedge funds, but the smaller universe of daily reporting Equity Long/Short funds points to a less close relationship with semiconductor stocks (Figure 1), suggesting that equity-focused hedge funds might have deleveraged/reduced their exposure to semiconductors during July.

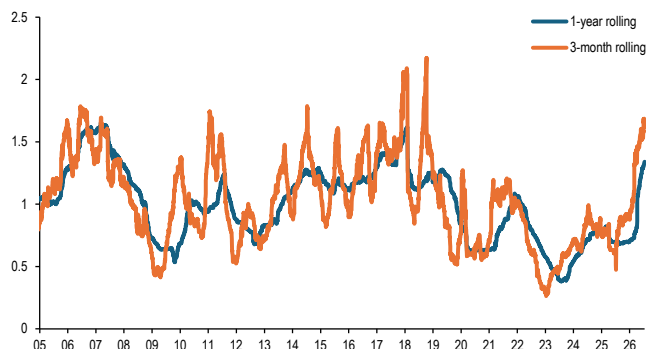
Figure 1: HFRX Equity Hedge index vs SMH Semiconductor equity ETF



Source: HFR, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- This is consistent with our higher-frequency leverage proxy for Equity Long/Short hedge funds, which as shown in Figure 2 it subsided somewhat in July after reaching its highest level since 2017 in June.

Figure 2: Our higher-frequency leverage proxy for equity Long/Short hedge funds

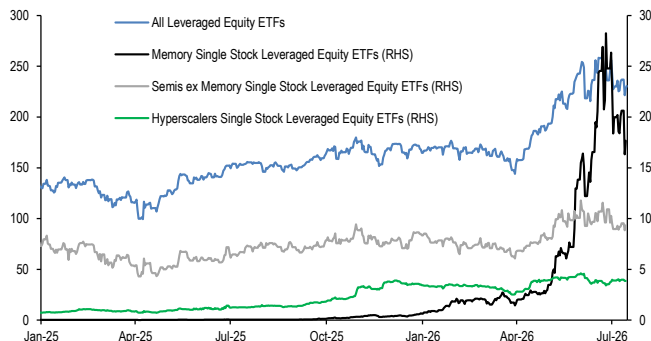


Source: HFR, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- In our opinion there is a need for normalizing in leverage across a wider investor spectrum beyond equity-focused hedge funds. Indeed, we see signs that this normalizing in leverage across a wide set of metrics beyond the HF leverage metric of Figure 2.
- For leveraged equity ETFs, volatile range-trading is naturally reducing their AUM due to convexity. For example, if the underlying index falls 10% one day and rises 11.1% the following day to reach its previous level, a 3x leveraged ETFs would be down 30% on the first day and up 33.3% in the second day, thus ending 7% lower from its previous level. In other words, the excessive growth of leveraged equity ETFs can be self-correcting due to convexity: if excess leverage creates higher frequency of VaR shocks, a volatile range-trading in equities would over time shrink their AUM.
- Indeed, after peaking in June, the AUM of leveraged memory stock ETFs has shrunk by 34% while that of all leveraged equity ETFs has shrunk by 13% (Figure 3). That said, relative to the market cap of underlying stocks, the decline looks more modest as shown in Figure 4 for both the AUM of leveraged memory stock ETFs and of all equity ETFs. In our mind it will perhaps take another three months or so of volatile range-trading in equities before the leveraged equity ETF AUM ratios to the market cap of underlying stocks return to pre-April levels for either memory stocks or all equity ETFs.

Figure 3: AUM of leveraged equity ETFs

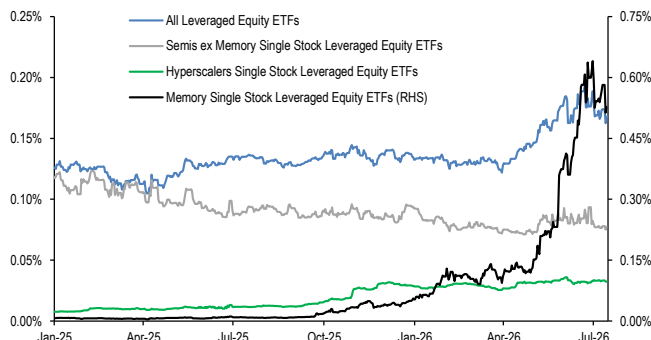
In \$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 4: AUM of leveraged equity ETFs as % of the market cap of underlying stocks

In %.

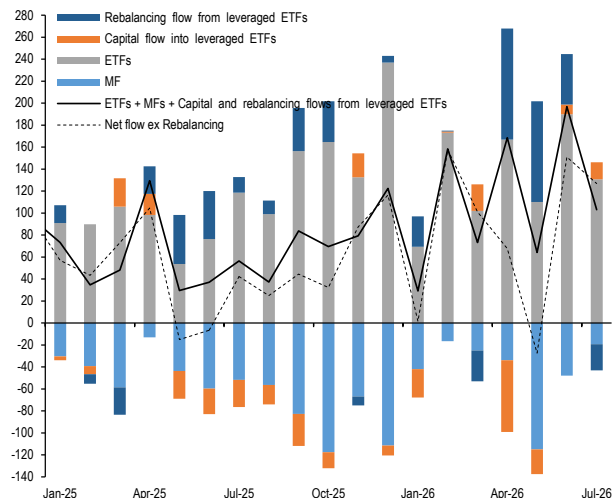


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- This is also because these leveraged equity ETFs saw further inflows in July as shown by the orange bars in Figure 5 and Figure 6, thus prolonging the duration of volatile range-trading needed to shrink/normalize their AUM.
- Finally, while Figure 4 illustrates why leveraged ETFs have become a bigger source of volatility for memory stocks, as their AUM ratio to the market cap of underlying stocks is three times higher for memory stocks than that of all equity ETFs, the elevated level of the dark blue line in Figure 4 relative to its own history reveals that the leveraged ETF problem as a source of volatility is not only confined to memory stocks but is a broader problem for the overall equity market due to the proliferation of leveraged equity index ETFs.

Figure 5: Flows stemming from equity funds globally

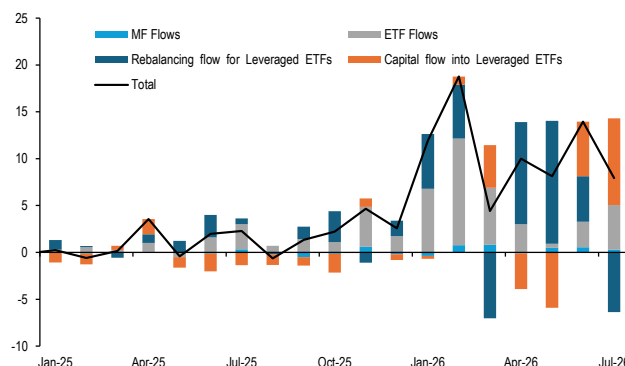
In \$bn per month, July MTD.



Source: LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 6: Flows stemming from equity funds that invest in Korea

In \$bn per month, July MTD.

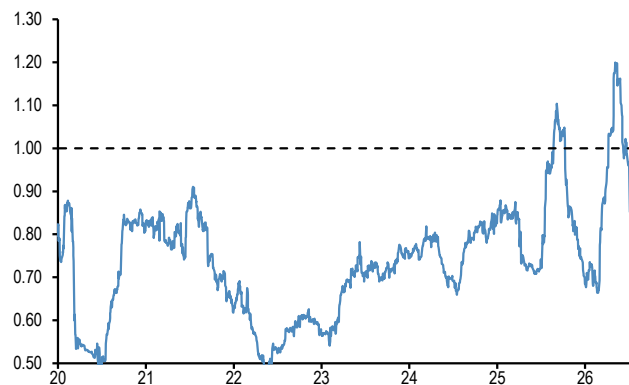


Source: LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- In the risk parity fund space, our leverage proxy has already normalized as shown in Figure 7, so excessive leverage by risk parity funds should no longer present a significant headwind for equities going forward.

Figure 7: Implied leverage of risk-parity funds

Ratio of 3-month rolling volatility of risk-parity funds' returns to the volatility of our risk parity strategy benchmark.

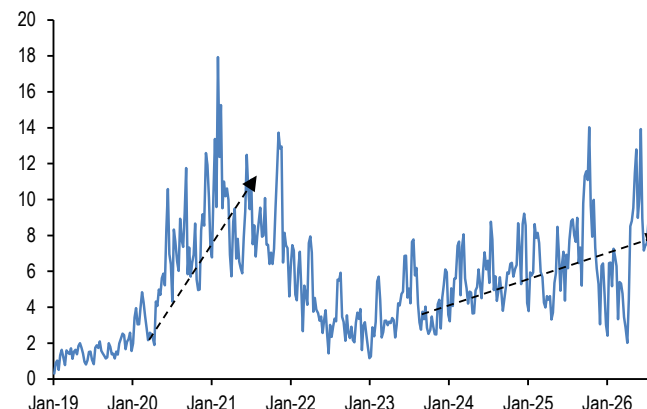


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- In the options space, previous extreme call option buying by retail investors appears to have subsided to a significant extent (Figure 8). As a reminder, our proxy of retail investors' option buying based on OCC data (on exchange-traded call option buying by customers with less than 10 contracts), retreated significantly after peaking on June 5th. The peak on June 5th at close to 14 million contracts had matched the previous peaks seen in October 2025 and November 2021. Following these previous peaks, tech stocks had seen a multi-month correction with the bottom/capitulation in tech stocks coinciding with low readings in this metric of between 2-4 million contracts. In other words, the peaking of this metric since the high of June 5th suggests the retail impulse in the options space is currently subsiding and thus could continue to present a headwind for tech stocks (the preferred habitat of retail investors) going forward if capitulation levels of 2-4 million contracts are eventually reached.

Figure 8: Exchange-traded Call Option Buys at Open minus Sells at Open for Customers with less than 10 contracts for options on individual equities

In mn contracts. Last obs is for the week ending 10th July 2026.

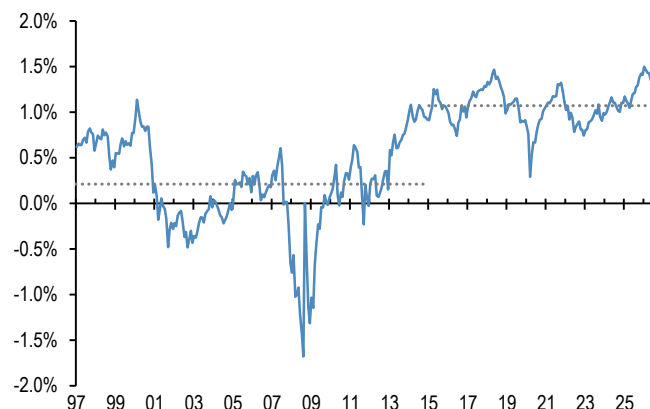


Source: OCC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Leverage in margin accounts remains rather elevated and thus significantly more deleveraging is needed in this space for margin accounts to no longer present a significant headwind for equities. As a reminder, for margin accounts we use the NYSE Net Debit balance as a proxy for US individual investor leverage. While in principle it includes both retail and institutional investors, we believe the majority is likely retail investors. In contrast to hedge funds, who can get leverage more easily or cheaply via options and futures, individuals rely more on Federal Reserve Regulation T. This regulation, which governs customer cash accounts and the amount of credit that brokerage firms and dealers may extend to customers for the purchase of securities, stipulates that one can borrow up to 50% of the purchase price of securities that can be purchased on margin. The NYSE Net Debit balance is calculated as the difference between margin debit balance minus the sum of total credit balances (cash accounts credit + margin account credit). This leverage proxy suggests that US individual investor leverage via margin accounts stands at extreme levels by historical standards, with tentative signs of retreating in recent months (Figure 9). This year's peak matches that seen at the end of 2021 or before then in mid 2018, with both of those previous episodes having been followed by a multi-month correction in equity markets.

Figure 9: Net Debit balances in NYSE margin accounts

The NYSE margin account Net Debit balance is equal to the margin debit balance minus the sum of total credit balances (cash account credit + margin account credit). The blue line is constructed by the Net Debit balance in \$bn divided by the market capitalisation of the S&P500 Index.



Source: FINRA, NYSE, J.P. Morgan Flows & Liquidity.

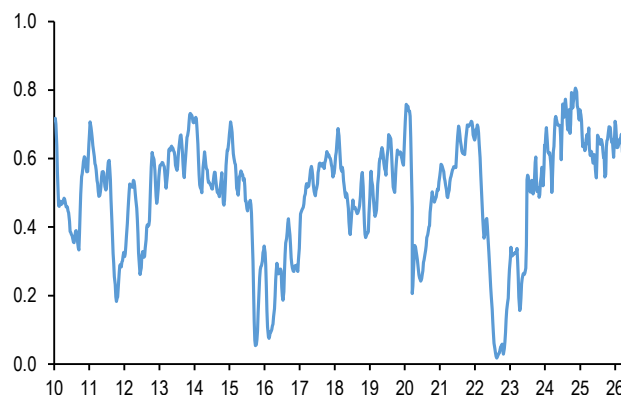
- In all, the investor deleveraging phase that started in June appears to be still ongoing and we see more room for deleveraging in leveraged equity ETFs, options and margin accounts, thus acting as a headwind for equities going forward. The excessive growth of leveraged equity ETFs can be self correcting: if excess leverage creates higher frequency of VaR shocks, a volatile range-trading in equities would over time shrink their AUM. That said, it will perhaps take another three months or so of volatile range-trading in equities until the leveraged equity ETF AUM ratios to the market cap of underlying stocks return to pre-April levels for either memory stocks or all equity ETFs.

Once this deleveraging abates, longer-term equity demand/supply dynamics should provide a support

- With global equities delivering a strong 10% return despite recent volatility, a question has arisen about which flows could support equities during the second half of the year.
- Starting with L/S hedge funds, the biggest equity hedge fund sector with an AUM of around \$1.4tr, we have typically proxied their equity beta by looking at the futures positions of asset managers and leveraged funds given equity L/S funds use futures as an overlay to achieve their desired beta. These futures positions of asset managers and leveraged funds are in turn proxied by the z-score of net CFTC positions as a % of open interest shown in Figure 10. A neutral z-score of zero is assumed to correspond to a historical average equity beta of 0.5 for Equity L/S hedge funds. We also assume that a very extreme 3 stdevs move in the z-score of CFTC futures positions corresponds to a very extreme 0.5 change in the equity beta. Using these assumptions, we estimate that the beta has declined somewhat from around 0.65 late last year ([F&L](#), Dec 3rd) and is around similar levels currently. This implies modestly positive equity demand of around \$20bn in 2026 to-date. As the above section notes, leverage remains elevated and we see little further increase in the beta into end-2026.

Figure 10: Proxy for Equity L/S positioning from US equity futures by Asset managers and Leveraged funds

CFTC positions in US equity futures by Leveraged funds and Asset managers (as a % of open interest). Rescaling as described in the text. It is an aggregate of the S&P500, Dow Jones, NASDAQ and their Mini futures contracts.



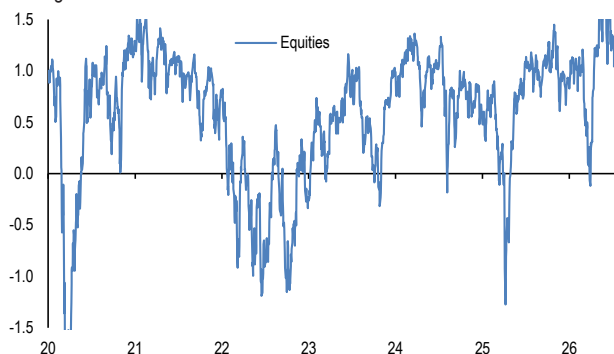
Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- For momentum-based investors such as CTAs, Figure 11 shows the average z-score of the short and long lookback period momentum signals for the S&P, Nikkei, Eurostoxx

50, FTSE 100 and MSCI EM indices. This z-score sat at around 0.9 at the turn of the year and is currently around 1.0 implying modest net purchases YTD of around \$20bn. For the remainder of the year, we continue to project a z-score of around 1.0 by end-2026, which in turn implies little net buying in 2H26.

Figure 11: Weighted average of the z scores of equity index momentum signals

Z-score of the momentum signals in our Trend Following Strategy framework shown in Tables A3 and A4 in the Appendix. The line shows the average z-score of the short and long lookback period momentum signals for the S&P 500, Nikkei, EuroStoxx50, FTSE100 and MSCI EM indices. We attach 50% weight on the S&P500 momentum signals and 50% weight on the momentum signals of the remaining four indices combined.

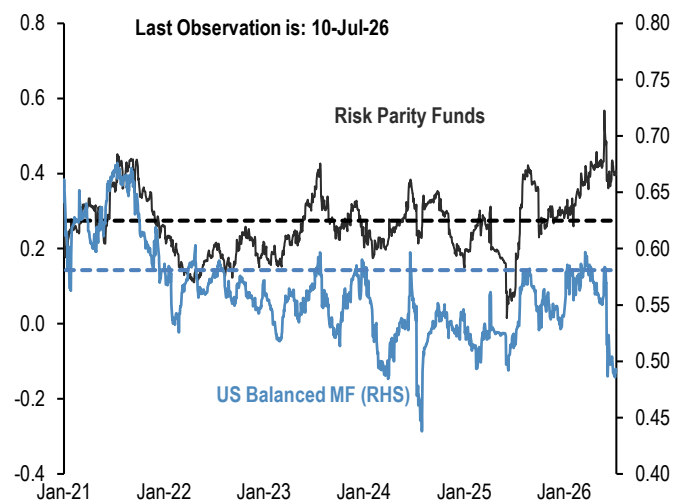


Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- What about the universe of risk parity funds? Estimates of its size vary, but a conservative estimate suggests an AUM of around \$150bn. Our calculations (Figure 12) suggest that the risk parity beta to equities has risen to around 0.4 currently, not far from its levels at the time we outlined our outlook for the 2026 equity supply-demand balance. Assuming some mean reversion to its long-run average, this would imply modest negative net demand of around \$20bn in 2H26. For US balanced mutual funds, the beta to equities has declined notably to around 0.49, compared to 0.54 around the turn of the year, implying negative net demand YTD. While this is consistent with rebalancing from equities to HG bonds in end-May, much of the move has taken place in June and coincided with a sharp drop in the bond-equity correlation. As a result, we look though part of the move suggesting balanced mutual funds have sold around \$210bn of equities YTD. Moreover, given Figure 12 suggests that there has been a structural shift lower in the equity beta in recent years relative to its long-term range, we assume it will remain little changed into end-2026.

Figure 12: Equity beta of US Balanced Mutual funds and Risk Parity funds

Rolling 42-day equity beta based on a bivariate regression of the daily returns of our Balanced Mutual fund and Risk Parity fund return indices to the daily returns of the S&P 500 and BarCap US Agg indices. For further details, see Appendix Chart A21. Dotted lines are average since 2015.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

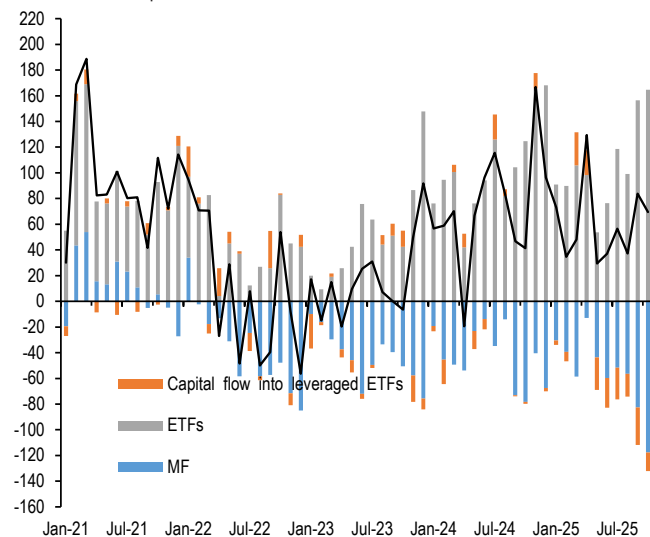
- What about pension funds and insurance companies? G4 insurance companies and pension funds, including both defined benefit and defined contribution plans, have typically been steady sellers of equities due to their structural shift away from equities towards fixed income. In 2025, the updated flow of funds data along with revisions to prior releases now suggests they sold around \$240bn of equities in 2025 and \$570bn in 2024. Late last year, we had projected net equity selling to reach similar levels to 2024 in the expectation of some pending rebalancing demand for this year, but similar to our mid-year update for the global bond supply-demand balance (F&L, Jun 24th) we now assume somewhat less pending rebalancing demand given the strength of bond purchases in 2H25 and project around \$470bn of net selling for 2026. Given we are broadly mid-way through the year, this leaves around \$235bn of net selling in 2H26.
- For SWFs and central banks, as we had noted previously (F&L, May 27th) there have been some questions over how the Middle East conflict would impact the recycling of current account surpluses. In practice, the impact has not been uniform, with some countries being able to continue exporting through alternative routes, and the period after the MoU had been signed had seen some resumption of traffic. And producers outside the region have benefited from higher revenues. In late 2025, we had projected around \$85bn of equity demand for 2026. With our commodity strategists now expecting a higher average oil price and with somewhat larger current account surpluses

for many oil producers, we now project equity demand of around \$110bn for 2026 of which around half should take place over 2H26.

- What about retail investor demand? In our 2026 equity supply-demand outlook, we had projected net demand from retail investors of around \$890bn using a forecasting model that projects the current year's equity flow as a % of AUM as a function of the previous year's equity return as well as previous year's bond return and flow as a % of AUM. YTD inflows have remained very strong (Figure 13) and are tracking if anything a stronger annualized pace of just over \$1.0tr. With net inflows into equity funds standing at around \$550bn in mid-July, this would suggest continued inflows of around \$480bn for the remainder of the year.

Figure 13: Overall equity fund flows including Mutual Funds, ETFs and Capital flows from Leveraged equity funds

\$bn. Jul 26 obs. up to 14th Jul.

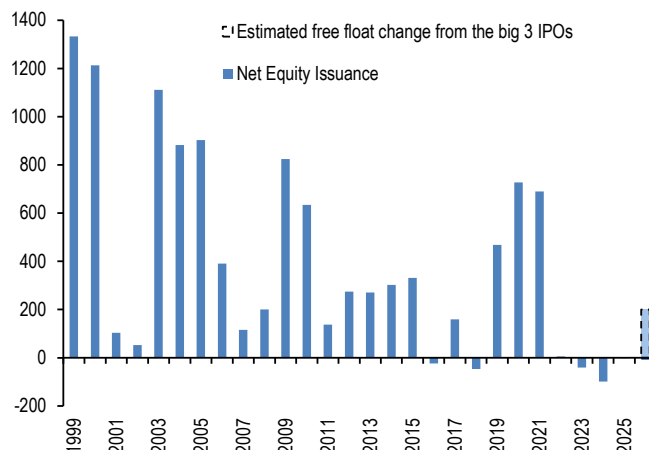


Source: LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- In terms of net supply, we had projected in late 2026 that net equity supply would remain broadly flat for 2026, similar to 2025. However, with the big 3 AI-related IPOs and recent announcements of plans to raise equity capital, a shift toward modestly positive net supply of around \$200bn for 2026 (Figure 14), of which perhaps around \$110bn has taken place. This suggests around \$90bn of further supply for the remainder of the year.

Figure 14: Net equity supply globally

\$bn per year based on the expansion of the MSCI AC World. Adjusted for price and FX changes.



Source: MSCI, J.P. Morgan Flows & Liquidity.

- Overall, this suggests equity demand for 2026 of around \$475bn, somewhat lower than the \$700bn we pencilled in late 2025, largely due to rebalancing by balanced mutual funds, while supply now looks higher than previously leaving the net demand at around \$275bn. We see around \$200bn of this net demand in 2H26.

Figure 15: Annual Global Equity Demand/Supply Balance

Flows per year in \$bn.

Year	2022	2023	2024	2025	2026	1H26	2H26
Demand							
Retail investors	11	228	913	757	1030	548	482
CTAs	-84	100	10	32	18	18	0
Equity L/S	-1191	835	489	-340	24	18	6
Risk Parity Funds	-26	47	-16	46	-27	-6	-21
Balanced MF	-386	189	-399	140	-210	-210	0
Pension & Insurance Funds	-179	-611	-570	-237	-470	-235	-235
SWF/Central banks	514	180	115	50	110	55	55
Total Demand	-1340	969	542	447	475	188	287
Supply	6	-41	-99	-9	200	110	90
Demand-Supply	-1346	1011	641	456	275	78	197

Source: J.P. Morgan Flows & Liquidity.

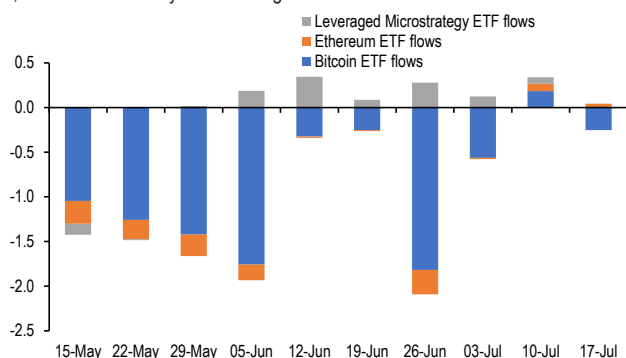
- Taken at face value, a positive equity demand/supply balance for H2 might appear at odds with the conclusion from our first section. However, we note a deleveraging process is likely to dominate in coming months, leading to significant swings in prices, while the equity demand-supply balance is more of a background / longer-term force that should provide some support once deleveraging abates.

Some encouraging signs from bitcoin futures

- While bitcoin ETF flows have been rather erratic in recent weeks, with inflows last week followed by outflows this week, the inflows into Leveraged MicroStrategy ETFs have been steadier and more positive over the past seven weeks (Figure 16).

Figure 16: Weekly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

\$bn, x-axis shows Friday week-ending dates



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- This buying by retail investors has likely supported MicroStrategy's stock price in recent weeks and prevented its common stock from trading at a discount to NAV. At the same time, MicroStrategy (Not Covered) continues to increase its dollar reserves from \$2.55bn last week to \$3bn this week, equivalent to 20 months of dividends.
- This is an encouraging sign for the bitcoin outlook, in our view. We had noted previously that to reduce concerns that it would have to sell bitcoin reserves in the future to pay dividends, MicroStrategy's could opt to rebuild its dollar reserves to cover 2-3 years of dividends.
- Whether this build-up of dollar reserves by MicroStrategy has improved sentiment among bitcoin investors is difficult to say. That said, we find it encouraging that this week, in contrast to outflows from bitcoin ETFs, there has been a positive flow impulse from bitcoin futures, both CME futures (Figure 17) and perpetual futures (Figure 18), which tend to be more driven by institutional rather than retail investors.

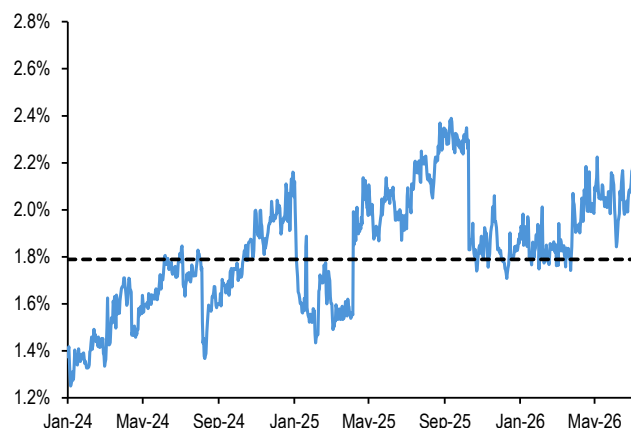
Figure 17: Our bitcoin position proxy, based on open interest in CME bitcoin futures contracts

In number of contracts. Last obs. for 14th July 2026.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 18: Ratio of open interest of bitcoin perps by bitcoin marketcap



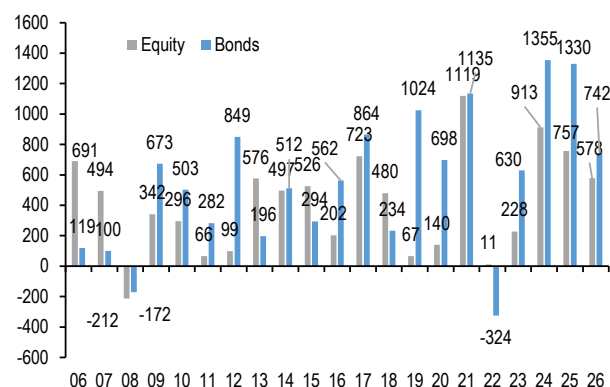
Source: Messari, Coin Metrics, J.P. Morgan Flows & Liquidity.

15 July 2026

Appendix

Chart A1a: Global equity & bond fund flows

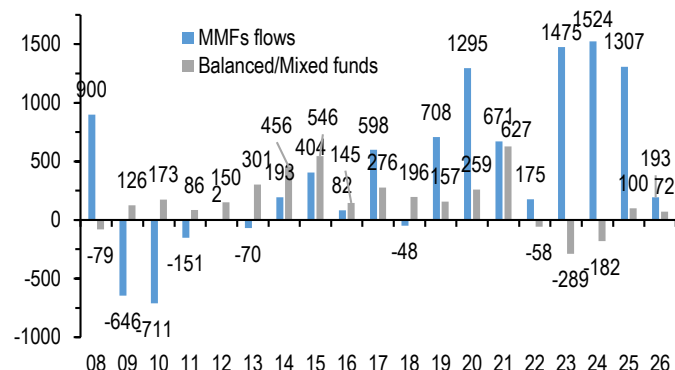
\$bn per year of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Flows come from ICI (worldwide data up to Q1'26). Data since then are a combination of monthly and weekly data from Lipper and Bloomberg.



Source: ICI, LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A1b: Quarterly Balanced/Mixed funds & MMFs flows worldwide

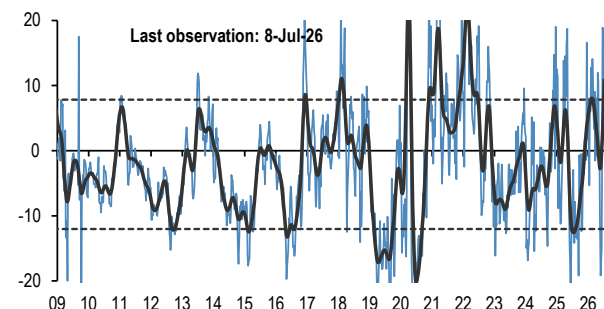
\$bn per quarter of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Data come from ICI (worldwide data) and are till Q1'26.



Source: ICI, J.P. Morgan Flows & Liquidity.

Chart A2: Fund flow indicator

Difference between flows into Equity and Bond funds: \$bn per week. Difference between flows into Equity vs. Bond funds in \$bn per week. Flows include Mutual Fund and ETF flows globally, i.e. funds domiciled both inside and outside the US (source: Lipper). The thin blue line shows the 4-week average of difference between Equity and Bond fund flows. Dotted lines depict ± 1 StDev of the blue line since 2009. The thick black line shows a smoothed version of the same series. The smoothing is done using a Hodrick-Prescott filter with a Lambda parameter of 100. China onshore A-share ETFs have been excluded.



Source: LSEG Lipper, J.P. Morgan Flows & Liquidity.

Table A2: Trading turnover monitor

Volumes are monthly and Turnover ratio is annualised (monthly trading volume annualised divided by the amount outstanding). UST Cash is primary dealer transactions in all US government securities. UST futures are from Bloomberg Finance L.P. JGBs are OTC volumes in all Japanese government securities. Bunds, Gold, Oil and Copper are futures. Gold includes Gold ETFs. Min-Max chart is based on Turnover ratio. For Bunds and Commodities, futures trading volumes are used while the outstanding amount is proxied by open interest. The diamond reflects the latest turnover observation. The thin blue line marks the distance between the min and max for the complete time series since Jan-2005 onwards. Y/Y change is change in YTD notional volumes over the same period last year.

As of Apr-26	MIN	MAX	Turnover ratio	Vol (tr)	y/y chng
Equities					
EM Equity			1.1	\$1.7	118%
DM Equity			1.4	\$11.4	38%
Govt Bonds					
UST cash			12.9	\$20.3	1%
UST futures			0.9	\$13.8	3%
JGBs			47.2	¥4,982	8%
Bund futures			1.7	€7.5	12%
Credit					
US HG			1.2	\$0.8	11%
US HY			0.6	\$0.1	2%
US Convertibles			3.4	\$0.04	13%
Commodities					
Gold			65.4	\$1.5	67%
Oil			95.4	\$3.4	-5%
Copper			2.4	\$0.7	80%
Digital Assets					
CME Bitcoin			116.0	\$0.075	-34%
CME Ethereum			122.4	\$0.036	61%

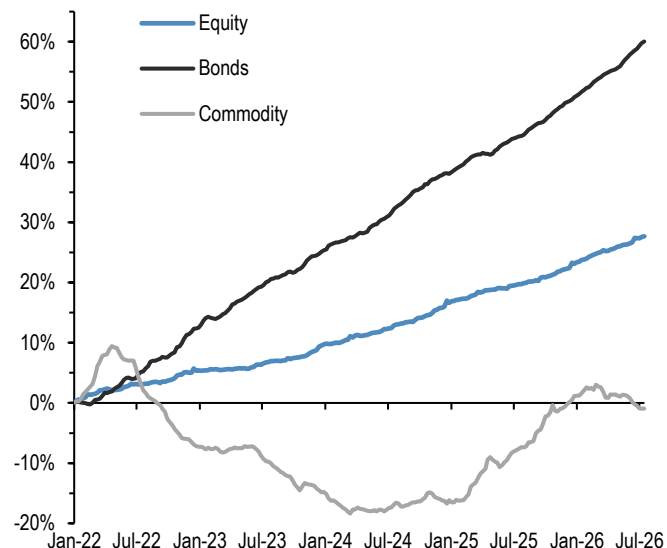
* Data with one month lag

Source: Bloomberg Finance L.P., Federal Reserve, Trace, Japan Securities Dealer Association, WFE, J.P. Morgan Flows & Liquidity.

ETF Flow Monitor (as of 15th July)

Chart A3: Global Cross Asset ETF Flows

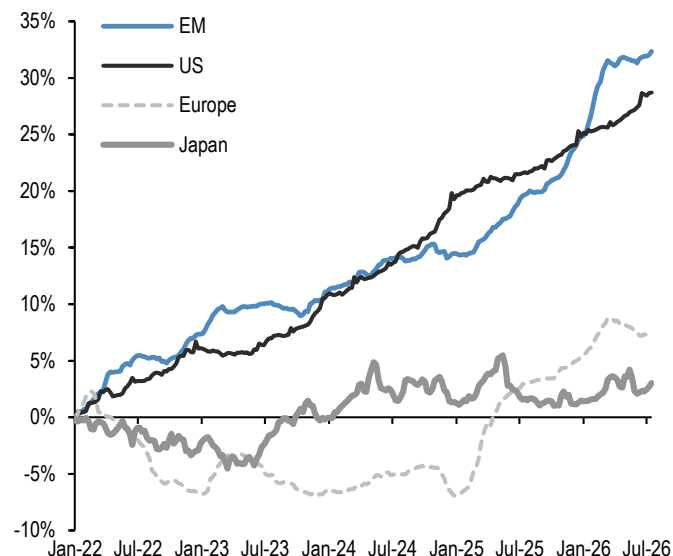
Cumulative flows into ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity. China onshore A-share ETFs have been excluded.

Chart A5: Global Equity ETF Flows

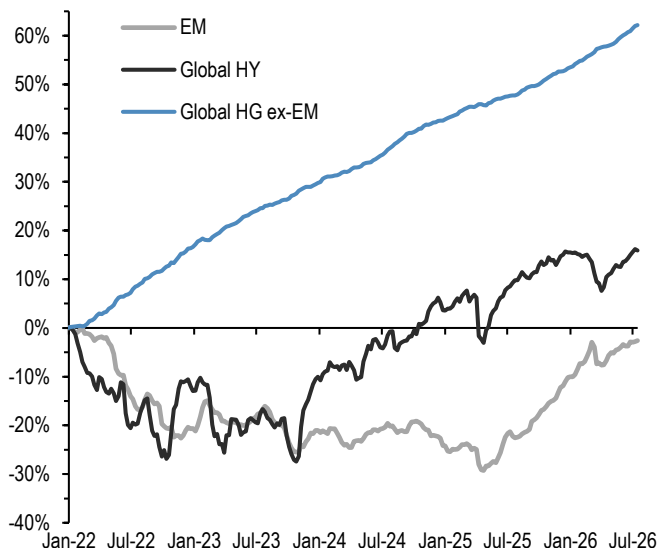
Cumulative flows into global equity ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.
 Note: We include ETFs with AUM > \$200mn in all the flow monitor charts. In Chart A5 for Japan, we subtract the BoJ buying of ETFs. China onshore A-share ETFs have been excluded.

Chart A4: Bond ETF Flows

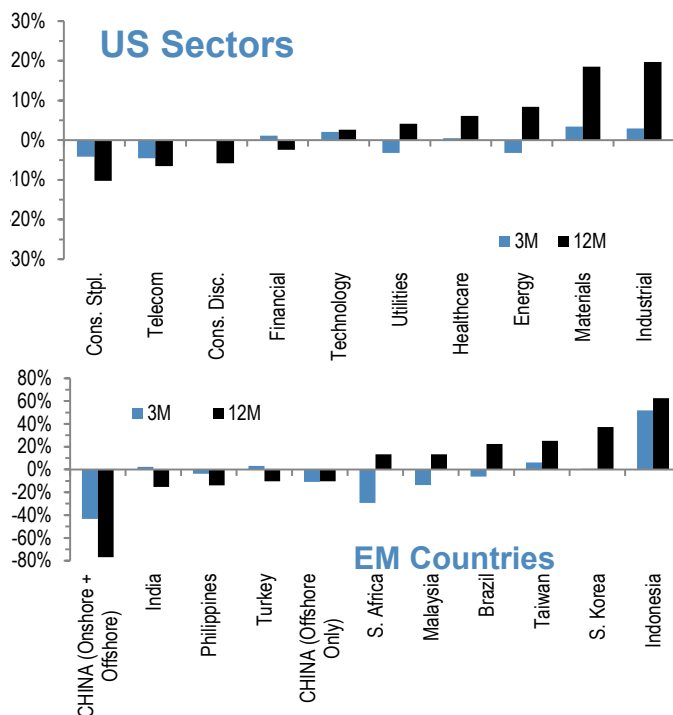
Cumulative flows into bond ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A6a: Equity Sectoral and Regional ETF Flows

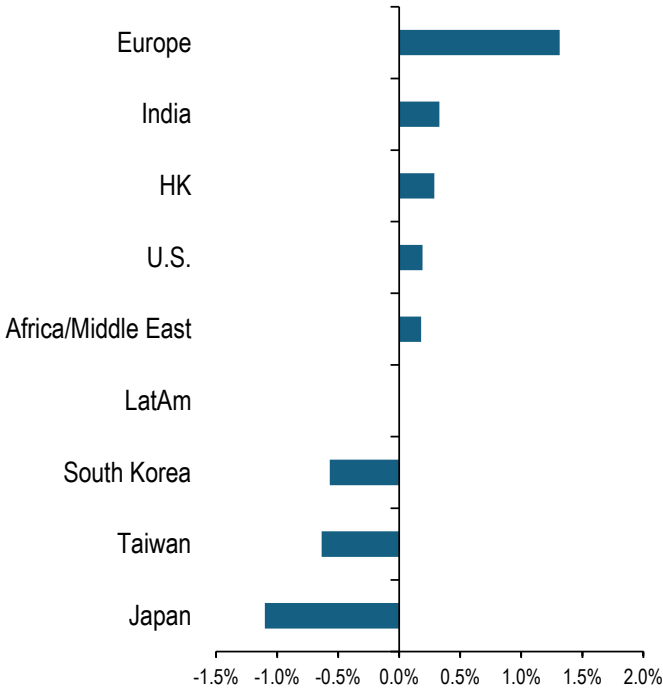
Rolling 3-month and 12-month change in cumulative flows as a % of AUM. Both sorted by 12-month change



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A6b: Regional Equity Allocation Monitor Implied by ETFs

In %.

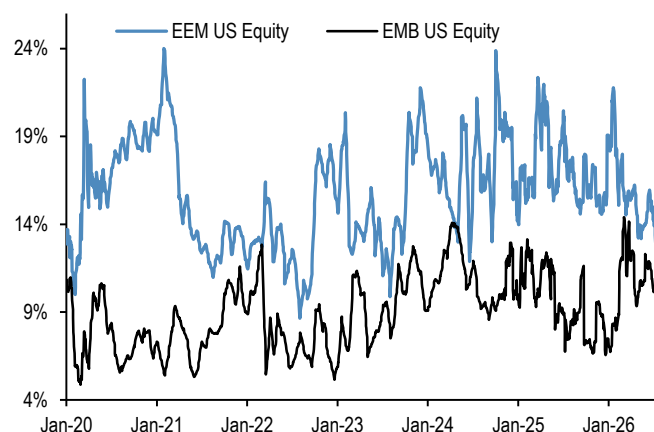


Source: Bloomberg Finance L.P., MSCI and J.P. Morgan Flows & Liquidity.
Note: Our regional equity allocation monitor implied by ETFs is based on a comparison of the share of each region in the equity ETF space relative to their corresponding shares in global equity indices.

Short Interest Monitor

Chart A7: Short interest on the EEM and EMB US ETF

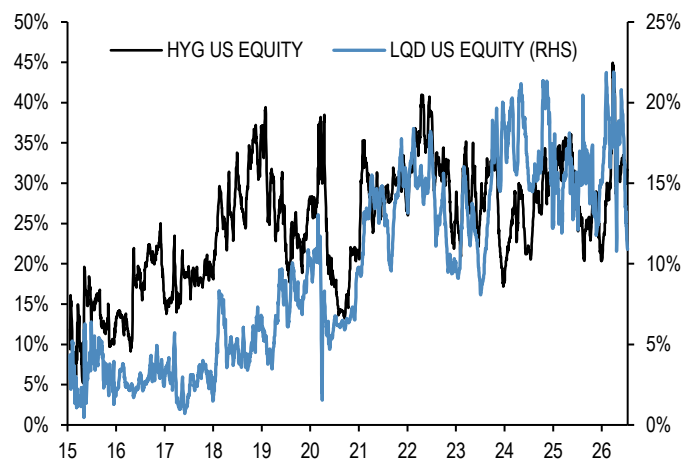
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A8: Short interest on the LQD and HYG US ETF

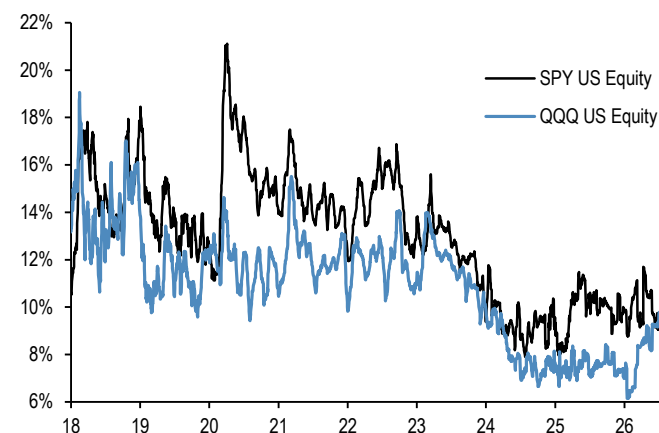
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A9: Short interest on the SPY and QQQ US ETF

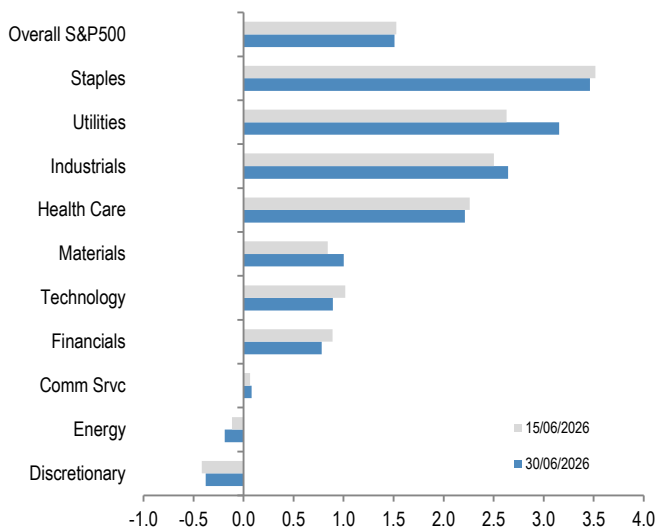
Short Interest as a % share of shares outstanding. Last obs is for 3rd July 2026.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A10: S&P500 sector short interest

Short interest as a % of shares outstanding based on z-scores. A strategy which overweights the S&P500 sectors with the highest short interest z-score (as % of shares o/s) vs. those with the lowest, produced an information ratio of 0.7 with a success rate of 56% (see F&L, Jun 28, 2013 for more details).



Source: NYSE, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A11a: Cross Asset Volatility Monitor 3m ATM Implied Volatility (1y history), as of 13th July-2026

This table shows the richness/cheapness of current three-month implied volatility levels (red dot) against their one-year historical range (thin blue bar) and the ratio to current realised volatility. Assets with implied volatility outside their 25th/75th percentile range (thick blue bar) are highlighted. The implied-to-realised volatility ratio uses 3-month implied volatilities and 1-month (around 21 trading days) realised volatilities for each asset.

Asset	Current	Low	Low date	High	High date		Upside	Downside	Implied/realized volatility
S&P 500	15%	13%	28-Aug-25	24%	27-Mar-26		9%	2%	1.13x
EuroSTOXX	16%	13%	18-Sep-25	26%	27-Mar-26		11%	2%	1.04x
Nikkei 225	31%	18%	22-Jul-25	33%	31-Mar-26		3%	13%	0.89x
Hang Seng	20%	17%	23-Dec-25	24%	26-Mar-26		4%	3%	0.97x
MSCI EM	34%	12%	15-Aug-25	38%	01-Jul-26		5%	21%	0.89x
Gold	23%	14%	19-Aug-25	33%	26-Mar-26		11%	9%	0.85x
Oil (brent)	51%	27%	19-Sep-25	89%	30-Mar-26		38%	25%	0.99x
Copper	18%	14%	18-Aug-25	37%	03-Feb-26		19%	3%	0.67x
BB commodity index	18%	13%	10-Dec-25	25%	30-Mar-26		7%	5%	1.07x
EUR/USD	8%	8%	10-Jul-26	13%	27-Mar-26		4%	0%	1.65x
USD/NOK	8%	8%	17-Dec-25	11%	30-Mar-26		3%	0%	0.83x
USD/JPY	7%	7%	29-May-26	11%	15-Jul-25		3%	0%	1.43x
GBP/USD	6%	6%	10-Jul-26	8%	27-Mar-26		2%	0%	1.01x
USD/CHF	7%	6%	29-May-26	9%	29-Jan-26		2%	1%	1.00x
10y US swaps	7%	6%	29-May-26	10%	27-Mar-26		3%	1%	1.04x
10y Eur swaps	58%	43%	26-Feb-26	97%	27-Mar-26		39%	15%	1.02x
CDX IG	43%	38%	29-Jul-25	66%	27-Mar-26		23%	5%	1.97x
CDX HY	40%	31%	09-Jan-26	57%	30-Mar-26		17%	9%	1.58x
iTraxx	43%	37%	09-Jan-26	70%	27-Mar-26		27%	5%	1.61x
iTraxx X/O	21%	19%	23-Dec-25	26%	17-Oct-25		4%	2%	0.80x

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

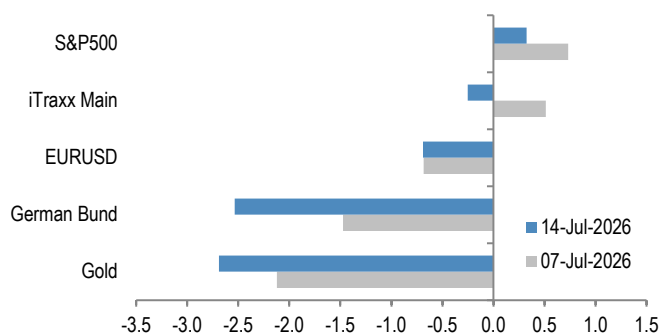
Note: Swaps volatility is 3m 10y payer ATM implied annualized BP vol and credit volatility is 3m 5y on-the-run ATM spread volatility. MSCI EM, Gold, Oil, Copper, BB Commodity Index and Treasury futures are 3m implied vol from Bloomberg.

Definitions:

Current:	Latest available closing level (13-Jul-26)
Low:	Lowest closing level in the last 1y
Low date:	Date the lowest closing level was reached (or the first time it was reached in the case of several identical low closing levels)
High:	Highest closing level in the last 1y
High date:	Date the highest closing level was reached (or the first time it was reached in the case of several identical high closing levels)
Graph:	Shows the current level and the 25th/75th percentile relative to the 1y high/low
Upside:	Implied return/volatility percentage points from current level up to the High (note: return is calculated as simple difference for spread products)
Upside (σ):	Upside in terms of standard deviations (Upside / Current 1y realized volatility)
Downside:	Implied return/volatility percentage points from current level down to the Low (note: return calculated as simple difference for spread products)
Downside (σ):	Downside in terms of standard deviations (Downside / Current 1y realized volatility)
Implied/realized volatility:	Current 3m implied volatility / current realized 3m volatility

Chart A11b: Option skew monitor

Skew is the difference between the implied volatility of out-of-the-money (OTM) call options and put options. A positive skew implies more demand for calls than puts and a negative skew, higher demand for puts than calls. It can therefore be seen as an indicator of risk perception in that a highly negative skew in equities is indicative of a bearish view. The chart shows z-score of the skew, i.e. the skew minus a rolling 2-year avg skew divided by a rolling two-year standard deviation of the skew. A negative skew on iTraxx Main means investors favour buying protection, i.e. a short risk position. A positive skew for the Bund reflects a long duration view, also a short risk position.

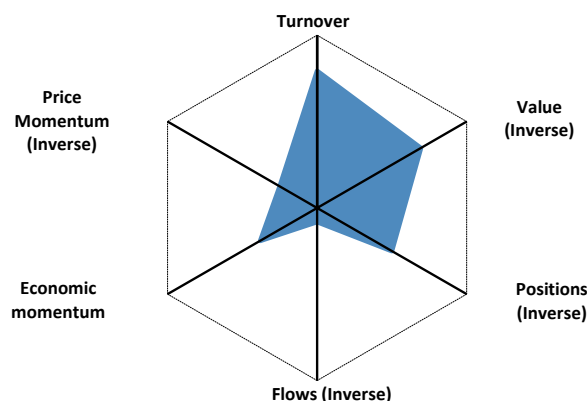


Source: J.P. Morgan Flows & Liquidity.

Chart A11c: Equity market health map

Each of the six axes corresponds to a key indicator for markets. The position of the blue line on each axis shows how far the current observation is from the extremes at either end of the scale. Overall, the larger the blue area within the hexagon, the better the outlook for the S&P500. All variables are expressed as the percentile of the distribution that the observation falls into. I.e. a reading in the middle of the axis means that the observation falls exactly at the median of all historical observations.

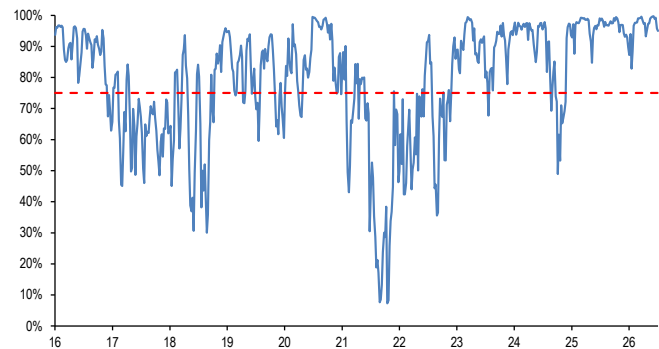
Turnover: Percentile rank of the 21-day rolling S&P500 turnover since 2007. **Value:** Inverse of percentile rank of our quarterly S&P 500 Equity Risk Premium (ERP) estimate since 1957 as described in "A fair value model for US Bonds, Credit and Equities", Panigirtzoglou and Loeys, Jun 2005. **Positions:** Percentile rank of the difference between government bonds and equity positioning from the cross-asset positioning monitor on the front page of our publication since 2016. **Flows:** Inverse of percentile rank of the difference between equity and bond fund flows from the cross-asset fund flow monitor on the front page of our publication since 2007. **Economic momentum:** Percentile rank of the 2-month change in the global manufacturing PMI since 1998, see "REVISITING: Using the Global PMI as trading signal", Nikolaos Panigirtzoglou, Jan 2012). **Price momentum:** Inverse of the percentile rank of the z-score of the difference in the momentum signal of S&P500 vs. 10y USTs since 2016 (based on the average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 in the Appendix). China onshore A-share ETFs have been excluded. Last obs. as of 10th July.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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Chart A11d: Implied probability of the S&P500 rising over the next six months with a cutoff set at 75%



Source: J.P. Morgan Flows & Liquidity.

Chart A11e: Logistic regression model for direction of the S&P 500 over a 6 month horizon & Ex-ante probability of an up-move from the logistic regression algorithm vs. the ex-post realised 6-month return in the S&P 500

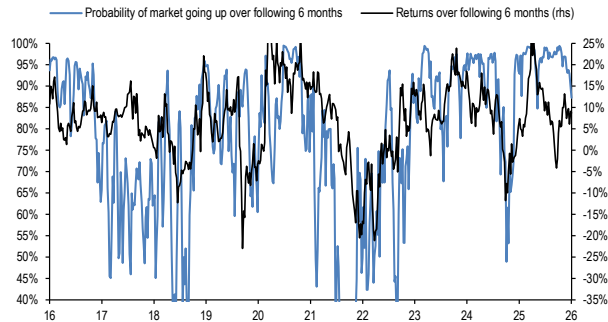
In %.

Dependent Variable: S&500 UP (1) or DOWN (0) over the next six months
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)

Sample (adjusted): 1/08/2016 12/20/2024
Included observations: 468 after adjustments

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	3.42	0.84	4.08	0.00
TURNOVER	2.97	0.72	4.15	0.00
VALUE	-3.47	0.90	-3.88	0.00
POSITIONS	-0.77	0.55	-1.39	0.16
FLows	-1.79	0.53	-3.38	0.00
ECON_MOMENTUM	3.35	0.62	5.42	0.00
MOMENTUM	-1.36	0.62	-2.19	0.03

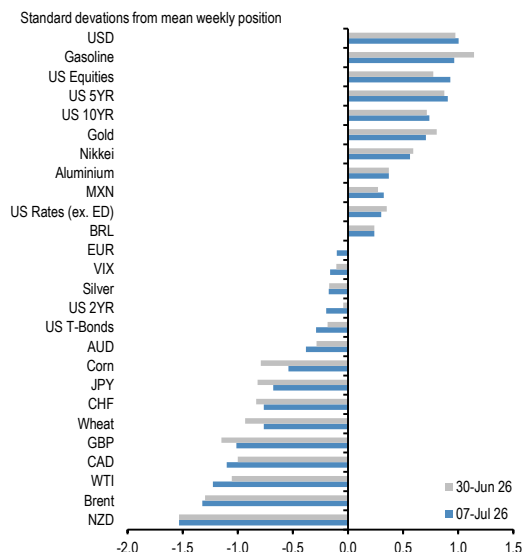


Source: J.P. Morgan Flows & Liquidity.

Spec position monitor

Chart A12: Weekly Spec Position Monitor

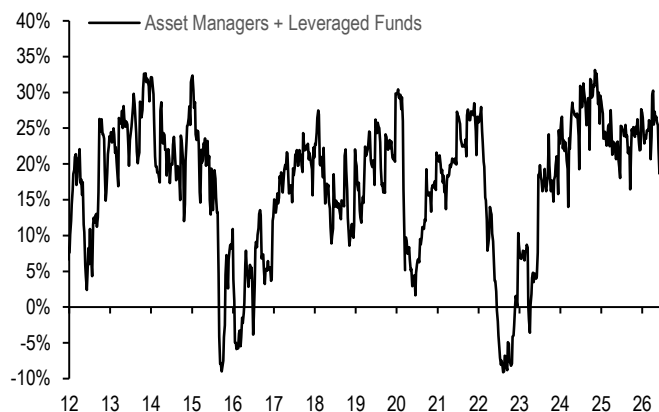
Net spec positions are proxied by the number of long contracts minus the number of short contracts using the speculative category of the Commitments of Traders reports (as reported by CFTC). To proxy for speculative investors for equity and US Treasury bond futures positions we use Asset managers and leveraged funds (see Chart A13), whereas for other assets we use the legacy Non-Commercial category. This net position is then converted to a dollar amount by multiplying by the contract size and then the corresponding futures price. We then scale the net positions by open interest. The chart shows the z-score of these net positions. US rates is a duration-weighted composite of the individual UST futures contracts excluding the Eurodollar contract.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A13: Positions in US equity futures by Asset managers and Leveraged funds

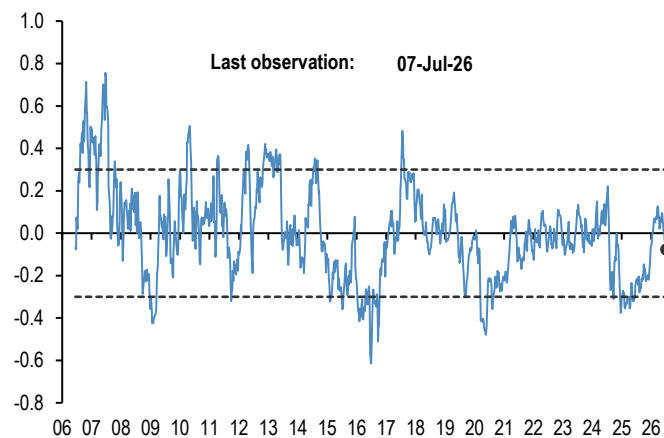
CFTC positions in US equity futures by Leveraged funds and Asset managers (as a % of open interest). It is an aggregate of the S&P500, DowJones, NASDAQ and their Mini futures contracts.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

Chart A14: Spec position indicator on Risky vs. Safe currencies

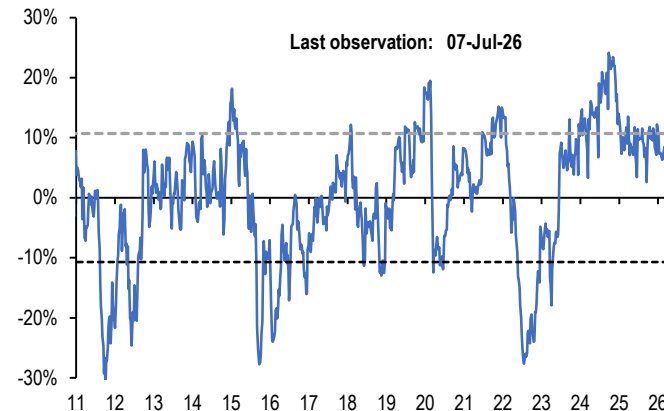
Difference between net spec positions on risky & safe currencies. Net spec position is calculated in USD across 5 'risky' and 3 'safe' currencies (safe currencies also include Gold). These positions are then scaled by open interest and we take an average of 'risky' and 'safe' assets to create two series. The chart is then simply the difference between the 'risky' and 'safe' series. The final series shown in the chart below is demeaned using data since 2006. The risky currencies are: AUD, NZD, CAD, MXN and BRL. The safe currencies are: JPY, CHF and Gold.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A15: Spec position indicator on US equity futures vs. intermediate sector UST futures

Difference between net spec positions on US equity futures vs. intermediate sector UST futures. This indicator is derived by the difference between total CFTC positions in US equity futures by Asset managers + Leveraged Funds scaled by open interest minus the Asset managers + Leveraged Funds spec position on intermediate sector UST futures (i.e. all UST futures duration weighted ex ED and ex 2Y UST futures) also scaled by open interest.



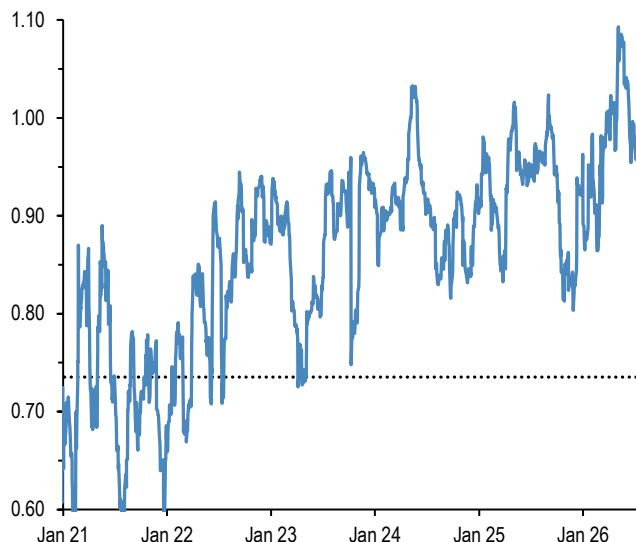
Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

15 July 2026

Mutual fund and hedge fund betas

Chart A16: 21-day rolling beta of 20 biggest active US bond mutual fund managers with respect to the US Agg Bond Index

The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A18: Performance of various types of investors

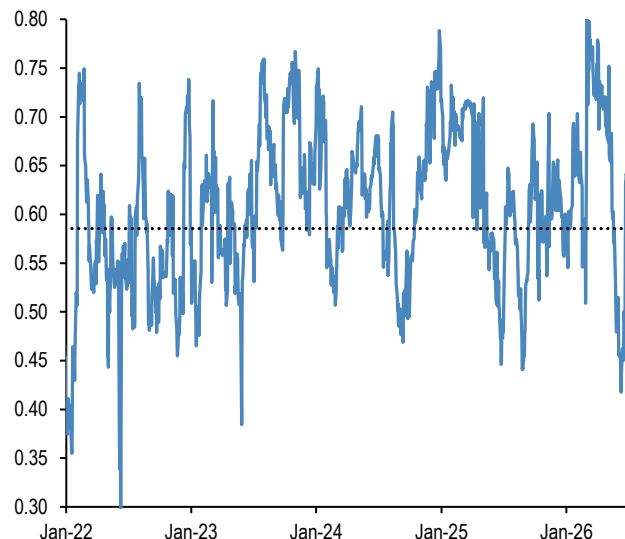
The table depicts the performance of various types of investors in % as of 14th July 2025.

Date	2020	2021	2022	2023	2024	2025	2026
Investors							
Equity L/S	13.4%	9.3%	-7.0%	11.3%	13.0%	15.5%	9.1%
Macro ex-CTAs	4.9%	4.3%	11.5%	1.3%	6.0%	10.2%	3.9%
CTAs	1.2%	8.8%	14.7%	-2.5%	2.9%	-1.1%	7.1%
Risk Parity Funds	3.5%	4.7%	-18.6%	6.0%	5.0%	17.3%	5.6%
US Balanced MFs	13.2%	14.4%	-13.0%	13.8%	11.4%	14.4%	6.6%
Benchmark							
MSCI AC World	16.3%	16.4%	-18.4%	22.2%	17.5%	22.3%	10.9%
Barclays Global Agg	5.6%	-2.5%	-11.2%	7.1%	3.4%	4.9%	0.4%
S&P Riskparity Vol 10	11.5%	12.8%	-16.2%	15.0%	5.4%	15.9%	13.4%
60 US Equity : 40 US Bonds	13.3%	14.8%	-15.4%	18.6%	16.4%	12.7%	6.6%

Source: Bloomberg Finance L.P., HFR, Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A17: 21-day rolling beta of 20 biggest active Euro bond mutual fund managers with respect to the Euro Agg Bond Index

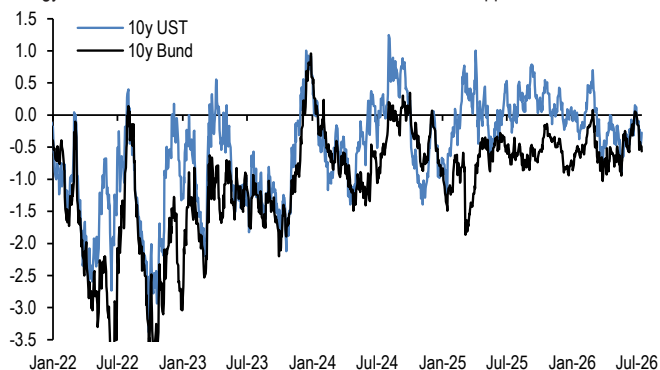
The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A19: Momentum signals for 10Y UST and 10Y Bunds

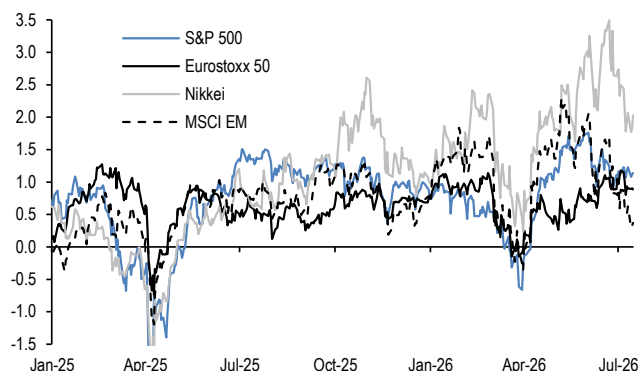
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A20: Momentum signals for S&P500

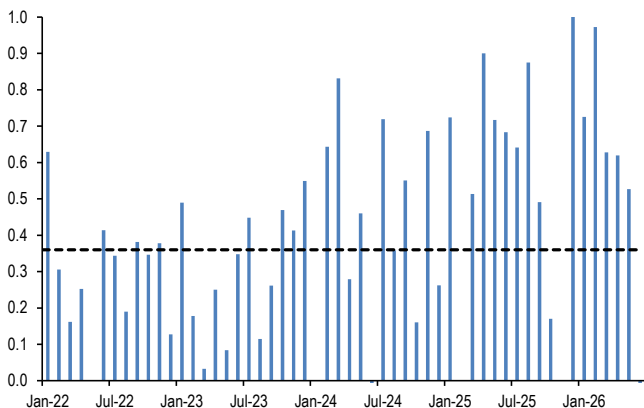
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A22: Equity beta of monthly reporting Equity Long/Short hedge funds

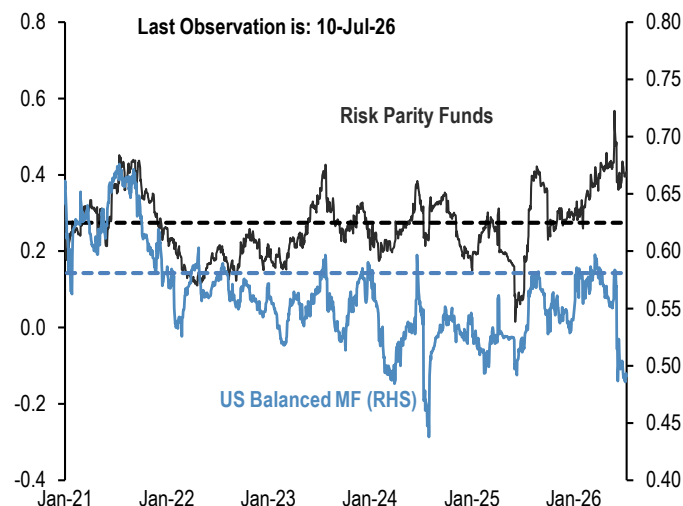
Proxied by the ratio of the monthly performance of Pivotal Path Asset-Weighted Equity Diversified hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. is June 26 based on provisional data.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A21: Equity beta of US Balanced Mutual funds and Risk Parity funds

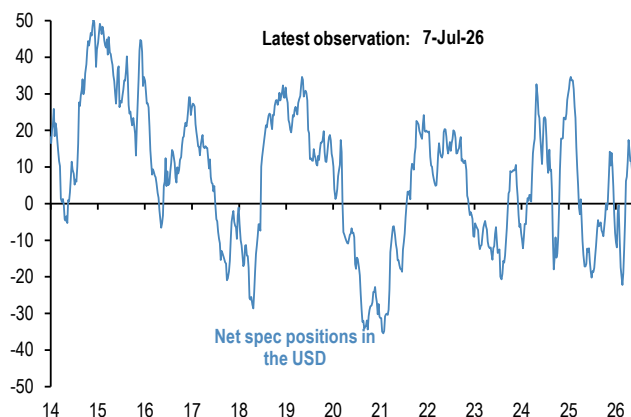
Rolling 42-day equity beta based on a bivariate regression of the daily returns of our Balanced Mutual fund and Risk Parity fund return indices to the daily returns of the S&P 500 and BarCap US Agg indices. Given that these funds invest in both equities and bonds we believe that the bivariate regression will be more suitable for these funds. Our risk parity index consists of 40 daily reporting Risk Parity funds. Our Balanced Mutual fund index includes the top 20 US-based active funds by assets and that have existed since 2006. Our Balanced Mutual fund index has a total AUM of \$700bn, which is around half of the total AUM of \$1.5tr of US based Balanced funds which we believe to be a good proxy of the overall industry. It excludes tracker funds and funds with a low tracking error. Dotted lines are average since 2015.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A23: USD exposure of currency hedge funds

The net spec position in the USD as reported by the CFTC. Spec is the non-commercial category from the CFTC.



Source: CFTC, Barclays, Datastream, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

CTAs – Trend following investors' momentum indicators

Table A3: Simple return momentum trading rules across various commodities

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg. days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
WTI	short	21				1	1	1.4	8.5%
	long	462	6.0	22.7	0.27	1	22	1.1	32.3%
Brent	short	84				-1	5	-0.5	-6.4%
	long	504	4.5	21.5	0.21	1	22	1.3	36.3%
Unleaded gas	short	105				1	34	0.8	11.4%
	long	483	3.1	23.2	0.14	0	11	2.1	56.4%
Heat Oil	short	63				1	4	1.2	12.9%
	long	483	4.6	22.0	0.21	0	95	3.0	92.4%
Gasoil	short	63				1	1	0.9	9.9%
	long	378	7.8	21.2	0.37	0	95	2.9	83.7%
Nat gas	short	105				-1	4	-0.6	-10.1%
	long	315	13.5	35.9	0.38	-1	116	-0.7	-21.6%
EU emission allowances	short	42				1	24	0.4	3.4%
	long	483	10.3	28.8	0.36	1	24	0.2	7.5%
1m fwd TTF Nat gas	short	21				0	1	2.0	19.6%
	long	294	36.3	31.4	1.16	1	81	1.0	46.0%
Gold	short	21				-1	9	-0.7	-1.9%
	long	483	3.2	10.6	0.30	1	27	0.5	6.9%
Silver	short	10				-1	2	-0.5	-1.9%
	long	462	6.3	19.7	0.32	1	18	0.6	13.7%
Palladium	short	42				0	0	0.0	0.0%
	long	273	12.0	23.1	0.52	-1	40	-0.5	-12.5%
Platinum	short	105				0	18	-1.7	-16.6%
	long	273	4.3	18.6	0.23	-1	18	-0.7	-10.9%
Aluminium	short	105				-1	16	-0.9	-6.8%
	long	357	5.7	15.2	0.38	1	21	0.7	9.4%
Copper	short	147				1	108	0.3	3.9%
	long	399	7.3	17.3	0.42	1	114	1.0	22.3%
Lead	short	126				-1	26	-0.6	-6.6%
	long	357	0.9	19.2	0.05	-1	127	-0.5	-11.0%
Nickel	short	42				-1	28	-0.7	-6.0%
	long	336	12.0	23.0	0.52	0	2	0.0	-0.2%
Zinc	short	126				1	76	0.5	6.0%
	long	399	7.9	19.6	0.40	1	206	0.8	19.0%
Wheat	short	168				1	9	0.7	8.3%
	long	294	4.2	23.2	0.18	1	9	0.6	8.9%
Kansas wheat	short	147				1	10	0.7	8.3%
	long	483	8.2	20.8	0.39	1	6	0.3	6.7%
Corn	short	63				-1	38	-0.4	-3.2%
	long	399	7.8	16.6	0.47	-1	50	-0.5	-10.0%
Soybeans	short	42				1	6	0.3	1.7%
	long	231	4.6	14.5	0.32	1	6	0.2	2.8%
Cotton	short	168				1	23	0.5	6.3%
	long	483	5.4	18.5	0.29	1	5	0.1	2.1%
Sugar	short	63				1	0	0.1	0.5%
	long	252	6.3	21.6	0.29	-1	43	-0.2	-3.9%
Coffee	short	63				0	3	2.0	17.1%
	long	273	7.9	24.0	0.33	1	6	0.3	6.0%
Cocoa*	short	10				1	2	0.8	3.2%

* For cocoa, uses only short-term momentum and a z-score threshold of 3 rather than 1.5 as for other contracts.

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Table A4: Simple return momentum trading rules across international equity indices, bond futures and FX

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

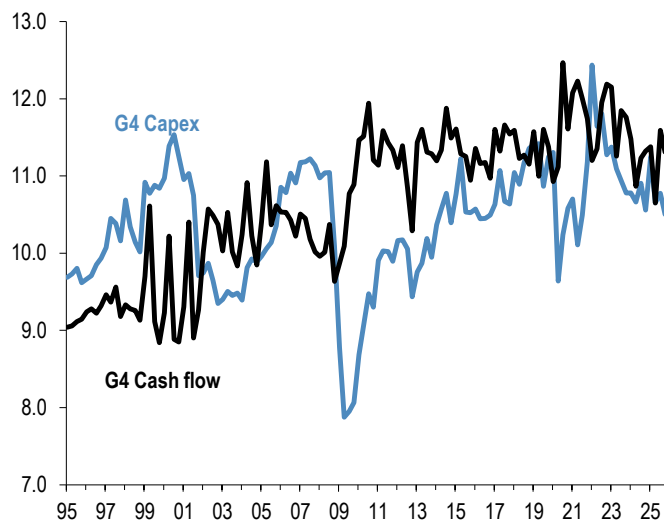
		Lookback (moving avg. days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
S&P 500	short	84	7.7	12.3	0.63	1	0	1.0	4.8%
	long	315				1	0	1.3	13.2%
Nasdaq 100	short	84	10.2	16.2	0.63	1	0	0.8	6.2%
	long	462				1	0	1.3	26.1%
Nikkei	short	63	4.1	13.3	0.31	1	0	0.8	4.5%
	long	420				0	0	2.9	44.5%
FTSE 100	short	168	5.3	12.1	0.44	1	0	0.6	3.8%
	long	504				0	0	1.6	17.8%
Eurostoxx 50	short	168	4.9	16.6	0.30	1	0	0.8	7.2%
	long	294				1	0	1.0	11.5%
MSCI EM	short	42	12.8	11.5	1.11	-1	0	-0.7	-3.4%
	long	357				1	0	1.3	21.2%
HYG credit	short	168	0.4	3.4	0.12	-1	0	-0.1	-0.4%
	long	273				-1	0	-0.1	-0.3%
LQD credit	short	10	2.6	6.5	0.40	0	0	0.0	0.0%
	long	189				-1	0	-0.1	-0.6%
2Y USTs	short	252	0.8	1.1	0.78	-1	0	-0.9	-1.1%
	long	462				-1	0	-0.6	-1.1%
5Y USTs	short	252	1.5	2.8	0.55	-1	0	-0.8	-1.9%
	long	420				-1	0	-0.5	-1.7%
10Y USTs	short	42	2.0	3.6	0.55	-1	0	-0.3	-0.5%
	long	441				-1	0	-0.4	-1.7%
2Y Schatz	short	189	0.6	0.8	0.81	-1	0	-0.8	-0.7%
	long	504				-1	0	-1.3	-2.3%
5y Bobl	short	63	1.3	1.9	0.71	-1	0	-0.4	-0.4%
	long	483				-1	0	-0.9	-2.9%
10y Bund	short	105	2.2	3.4	0.63	-1	0	-0.3	-0.7%
	long	462				-1	0	-0.6	-3.2%
10Y JGB	short	126	0.9	2.3	0.37	-1	0	-0.9	-0.9%
	long	273				0	0	-1.9	-2.8%
10Y Gilts	short	42	2.2	4.1	0.55	-1	0	-0.6	-0.9%
	long	399				-1	0	-0.8	-4.2%
10y OAT vs Bund	short	189	0.4	1.5	0.25	0	0	0.0	0.0%
	long	294				1	0	0.3	0.3%
10y BTP vs. Bund	short	84	2.2	6.3	0.36	1	0	0.1	0.3%
	long	273				1	0	0.1	0.5%
Euro	short	42	3.4	6.4	0.52	-1	0	-0.6	-1.2%
	long	273				-1	0	-0.8	-4.8%
Yen	short	21	3.7	6.3	0.58	-1	0	-0.4	-0.6%
	long	462				-1	0	-1.5	-12.2%
Sterling	short	168	1.3	7.1	0.19	-1	0	-0.4	-1.5%
	long	294				-1	0	-0.5	-2.5%
AUD	short	42	3.8	7.8	0.49	-1	0	-0.4	-1.1%
	long	420				1	0	0.2	2.0%
CAD	short	168	1.4	5.9	0.24	-1	0	-1.0	-3.5%
	long	504				-1	0	-1.0	-6.7%
G10 FX	short	42	2.1	5.2	0.42	1	0	0.5	0.8%
	long	189				1	0	0.6	2.1%
EMCI	short	42	2.6	5.0	0.52	-1	0	-0.2	-0.4%
	long	189				1	0	0.4	1.7%

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Corporate Activity

Chart A24: G4 non-financial corporate capex and cash flow as % of GDP

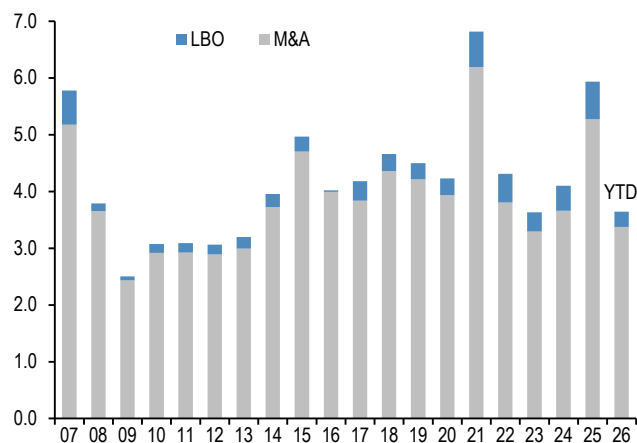
% of GDP, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A26: Global M&A and LBO

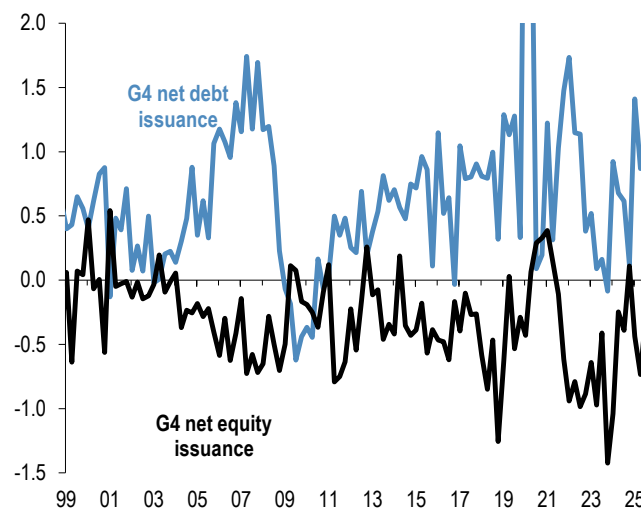
\$tr. M&A and LBOs are announced.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A25: G4 non-financial corporate sector net debt and equity issuance

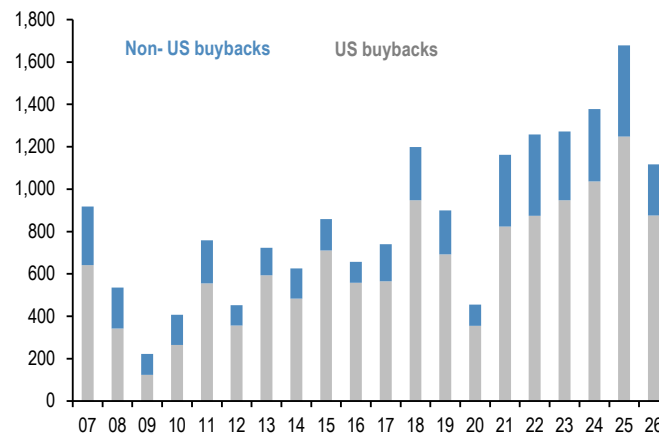
\$tr per quarter, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A27: US and non-US share buyback

\$bn, are as of June 26. Buybacks are as announced.



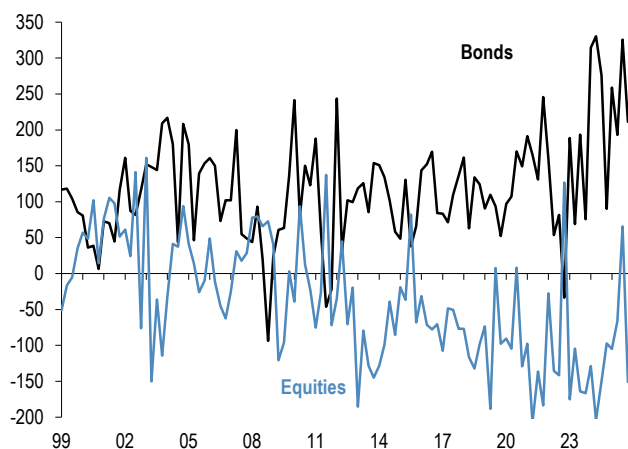
Source: Bloomberg Finance L.P., LSEG, J.P. Morgan Flows & Liquidity.

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Pension fund and insurance company flows

Chart A28: G4 pension funds and insurance companies equity and bond flows

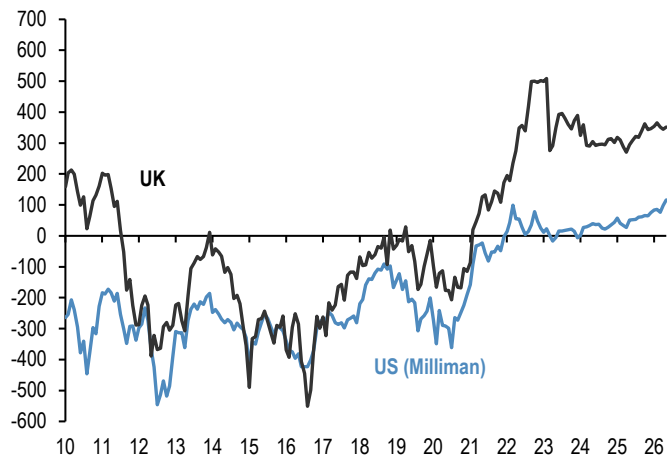
Equity and bond buying in \$bn per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A30: Pension fund deficits

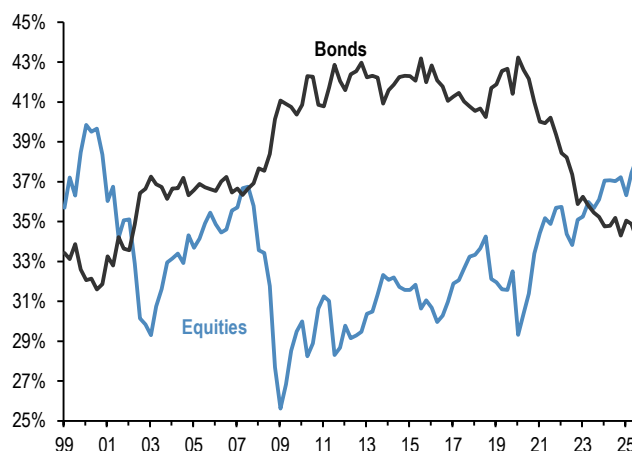
US\$bn. For US, funded status of the 100 largest corporate defined benefit pension plans, from Milliman. For UK, funded status of the defined benefit schemes eligible for entry to the Pension Protection Fund, converted to US\$ at today's exchange rates. Last obs. is May'26 for US & UK.



Source: Milliman, UK Pension Protection Fund, J.P. Morgan Flows & Liquidity.

Chart A29: G4 pension funds and insurance companies equity and bond levels

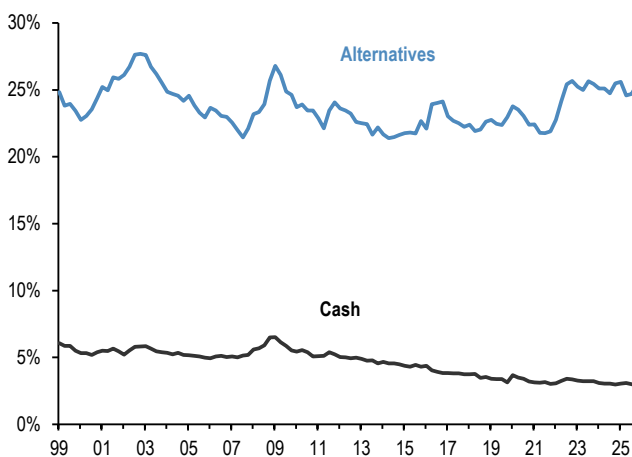
Equity and bond as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A31: G4 pension funds and insurance companies cash and alternatives levels

Cash and alternative investments as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.

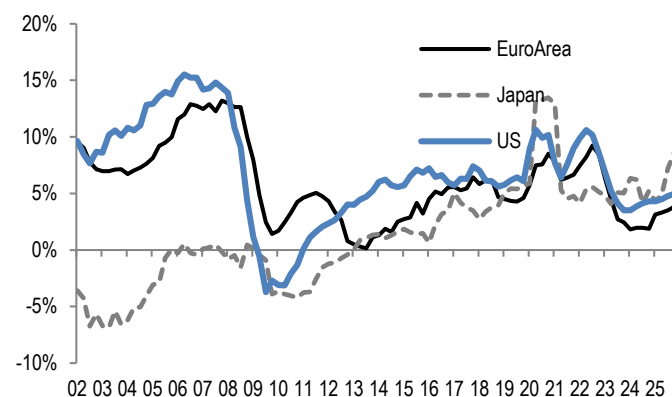


Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Credit Creation

Chart A32: Credit creation in the US, Japan and Euro area

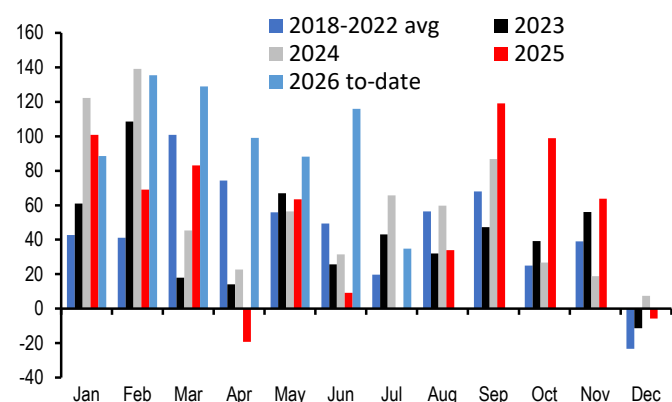
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. Last obs. is Q4'25 for Japan, Euro Area and US.



Source: Fed, ECB, BoJ, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A34: Monthly net issuance of US HG bonds

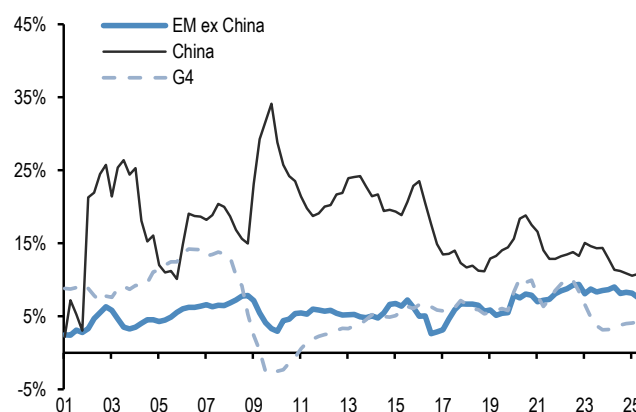
\$bn. Up to July 9th.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A33: Credit creation in EM

Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. The chart uses the FX rates as of the latest observation to calculate the credit creation at constant exchange rates. Last obs. is for Q2'25.



Source: G4 Central banks FoF, BIS, ICI, Barcap, Bloomberg Finance L.P., IMF, and J.P. Morgan Flows & Liquidity.

Table A5: Equity and Bond issuance

\$bn. Equity supply and corporate announcements are based on announced deals, not completed. M&A is announced deal value and buybacks are announced transactions. Y/Y change is change in YTD announcements over the same period last year.

Equity Supply	3-Jul	4 wk avg	13 wk avg	y/y chng
Global IPOs	5.4	26.9	12.2	216%
Secondary Offerings	13.3	11.7	13.0	28%
Corporate announcements				
M&A - Global	100.0	158.7	138.4	43%
- US Target	53.5	92.4	76.8	73%
- Non-US Target	46.5	66.3	61.6	21%
Net bond issuance	Feb-26	3 mth avg	YTD avg	y/y chng
USD	8	144	179	-50%
Non-USD	162	129	277	-4%

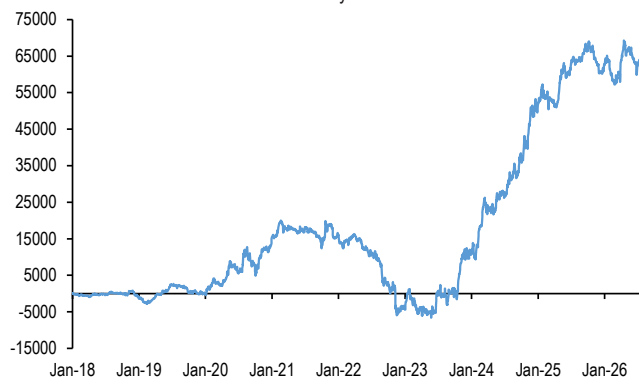
Source: Bloomberg Finance L.P., Dealogic, Thomson Reuters, J.P. Morgan Flows & Liquidity.

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Bitcoin monitor

Chart A35: Our Bitcoin position proxy, based on open interest in CME Bitcoin futures contracts

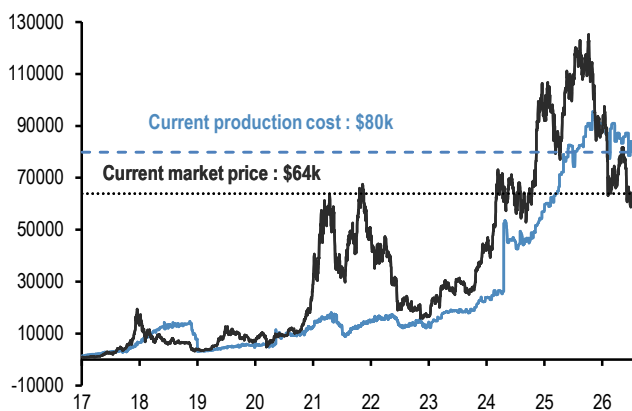
In number of contracts. Last obs. for 14th July 2026.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A37: Ratio of bitcoin market price to production cost

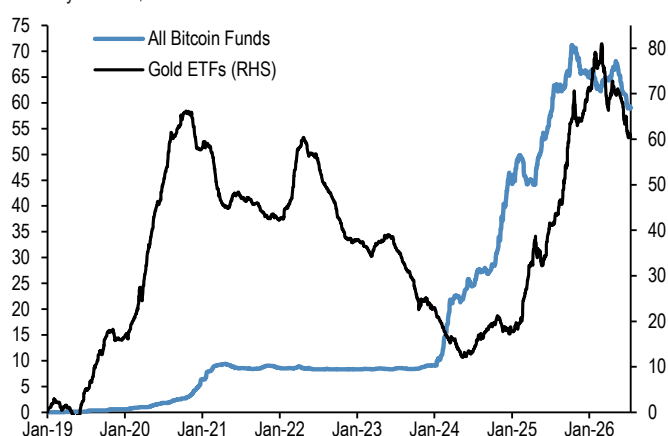
Based on the cost of production approach following Hayes (2018).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A36: Cumulative Flows in all Bitcoin funds and Gold ETF holdings

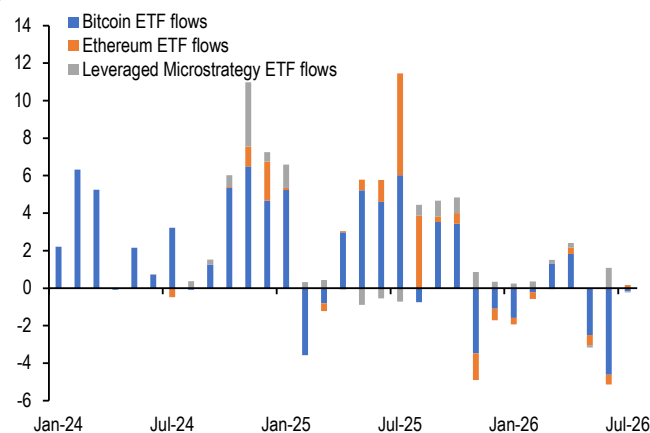
Both the y-axis in \$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A38: Monthly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

\$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Japanese flows and positions

Chart A39: Tokyo Stock Exchange margin trading: total buys minus total sells

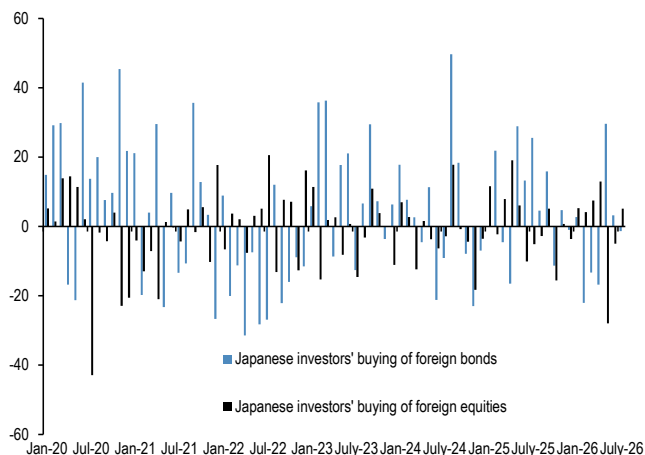
In bn of shares. Topix on right axis.



Source: Tokyo Stock Exchange, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A41: Monthly net purchases of foreign bonds and foreign equities by Japanese residents

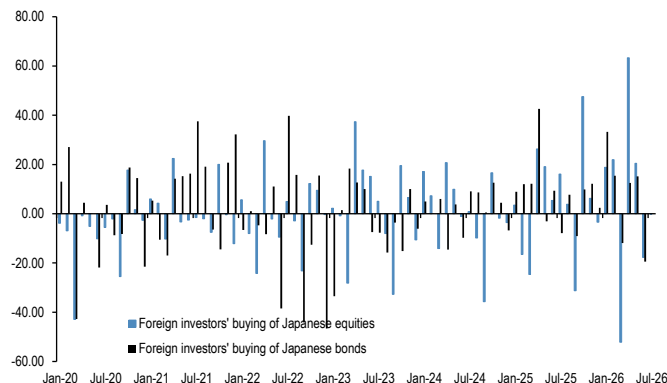
\$bn, Last weekly obs. is for 3rd July '26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A40: Monthly net purchases of Japanese bonds and Japanese equities by foreign residents

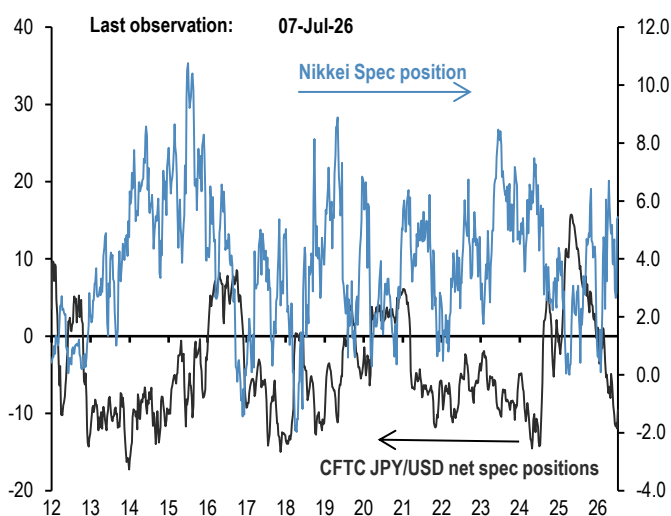
\$bn, Last weekly obs. is for 3rd July '26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A42: Overseas CFTC spec positions

CFTC spec positions are in \$bn. For Nikkei we use CFTC positions in Nikkei futures (USD & JPY) by Leveraged funds and Asset managers.



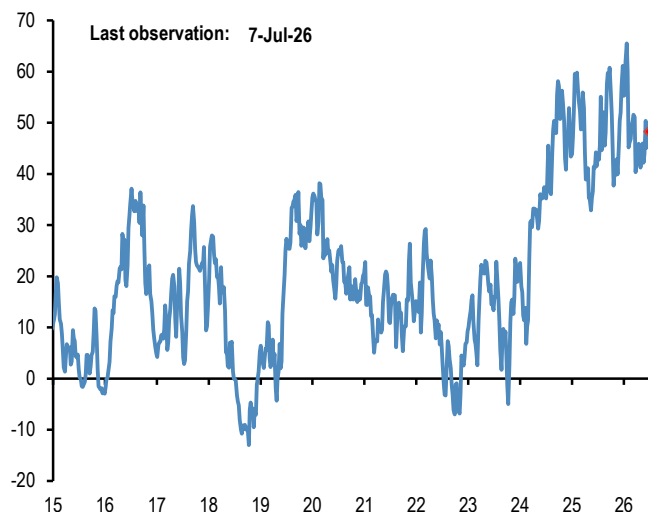
Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

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Commodity flows and positions

Chart A43: Gold spec positions

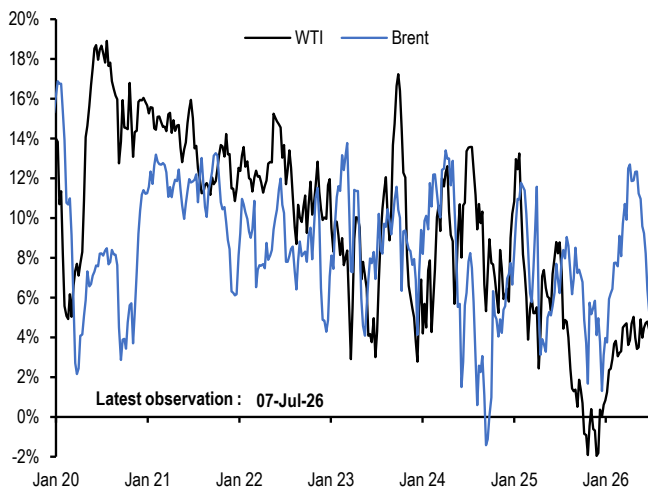
\$bn. CFTC net long minus short position in futures for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A45: Oil spec positions

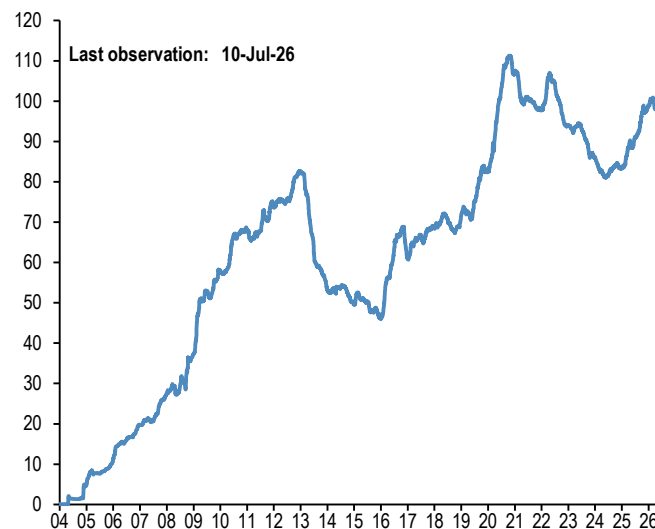
Net spec positions divided by open interest. CFTC futures positions for WTI and Brent are net long minus short for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A44: Gold ETFs

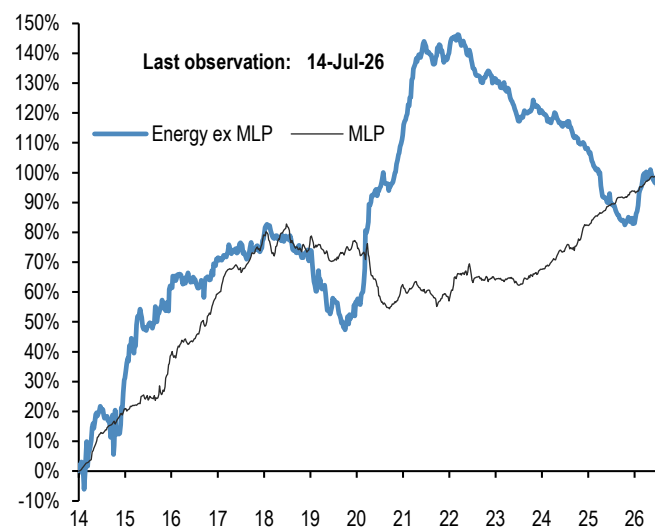
Mn troy oz. Physical gold held by all gold ETFs globally.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A46: Energy ETF flows

Cumulative energy ETFs flows as a % of AUM. MLP refers to the Alerian MLP ETF.

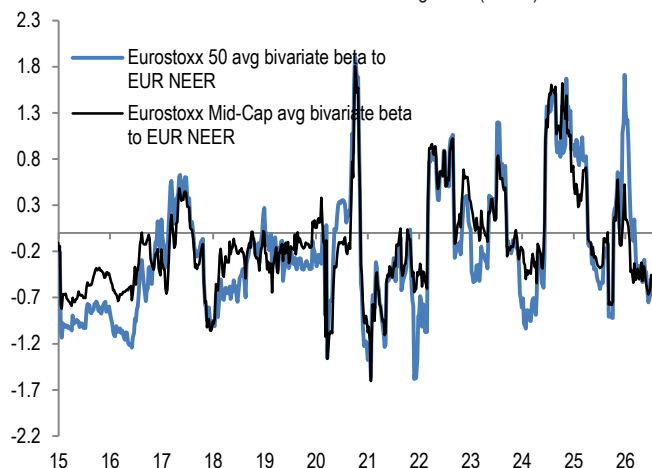


Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Corporate FX hedging proxies

Chart A47: Average beta of Eurostoxx 50 companies and Eurostoxx Mid-Cap to trade-weighted EUR

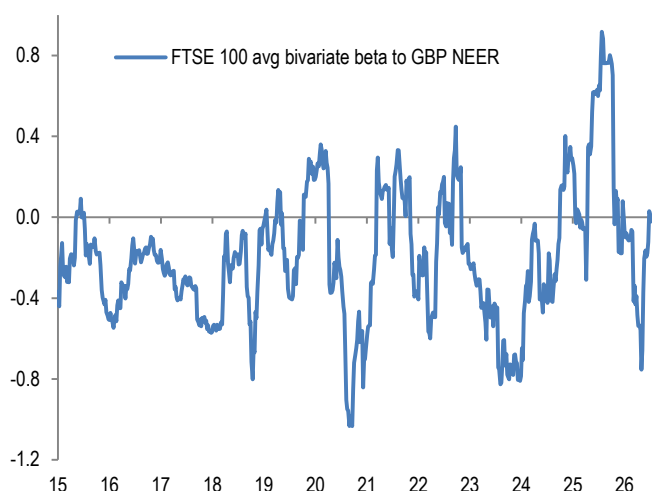
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the Eurostoxx 50 index to the weekly returns of the MSCI AC World and JPM EUR Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A49: Average beta of FTSE 100 companies to trade-weighted GBP

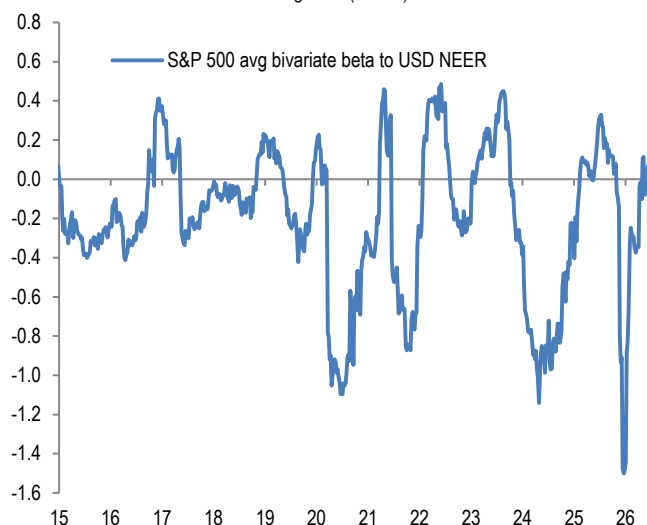
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the FTSE 100 index to the weekly returns of the MSCI AC World and JPM GBP Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A48: Average beta of S&P500 companies to trade-weighted US dollar

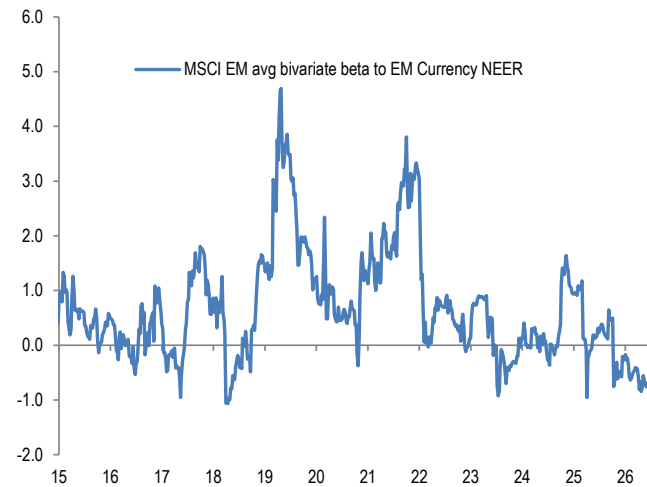
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of stocks in the S&P500 index to the weekly returns of the MSCI AC World and JPM USD Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A50: Average beta of MSCI EM companies to trade-weighted EM Currency Index

Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the MSCI EM index to the weekly returns of the MSCI AC World and JPM EM Nominal broad effective exchange rate (NEER).



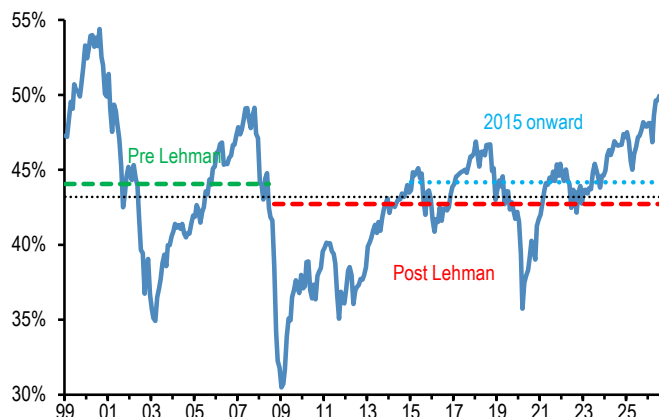
Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

15 July 2026

Non-Bank investors' implied allocations

Chart A51: Implied equity allocation by non-bank investors globally

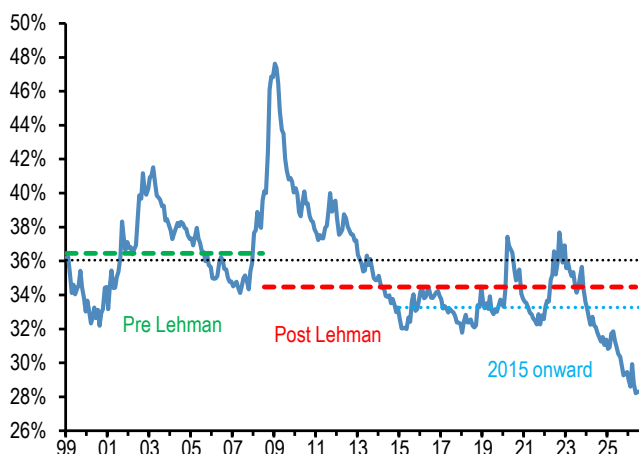
Global equities as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A53: Implied cash allocation by non-bank investors globally

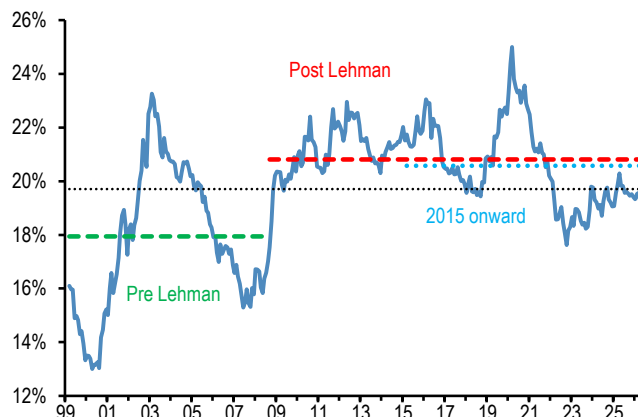
Global cash held by non-bank investors as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A52: Implied bond allocation by non-bank investors globally

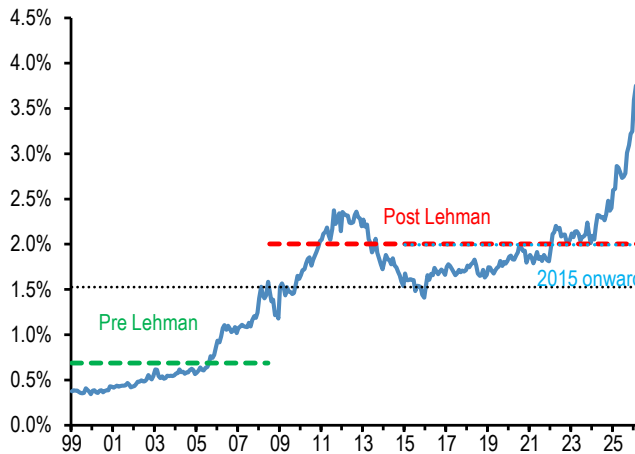
Global bonds as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A54: Implied commodity allocation (including privately held gold) by non-bank investors globally

Proxied by the open interest of commodity futures ex gold and privately held gold as % of the stock of equities, bonds, cash held by non-bank investors globally and commodities including gold held by private investors.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

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	Overweight (buy)	Neutral (hold)	Underweight (sell)
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